Influencer Dynamics*

Kazimier Smith†

November 10, 2026

Abstract

Consumers use social media for entertainment and to discover new products. To reach potential customers, brands pay influencers to feature products in their content. Payment depends on the size of the influencer's audience, and the effectiveness of the endorsement relies on trust. Excessive product recommendations may erode the relationship between an influencer and their followers. I develop a dynamic model in which an influencer produces sponsored posts (which include product recommendations) and organic posts (without product recommendations). Both affect the influencer's growth and require effort to produce. A sponsored post incurs additional costs: it requires searching for and negotiating with brands, and followers are less likely to engage with it. Using 2,780,011 Instagram posts and 136,453 TikTok posts from 1,369 influencers, I quantify the engagement penalty by comparing identical posts across platforms. Analyzing the influencers' career histories, I show that organic and sponsored posts have similar effects on follower growth. I leverage variation in the number and types of posts to estimate the unknown cost parameters in the model. An influencer with 100,000 followers optimally produces about 0.4 sponsored posts and 2.6 organic posts per week, and influencers with more followers produce more content of both types. Viral posts provide a transient boost to growth, but career progression largely depends on consistent content production. Regulating sponsored content can theoretically backfire by decreasing incentives to produce organic posts, but counterfactual simulations assuage this concern. A cost increase that reduces sponsored content by 25% only causes a 2.7% drop in organic content.

JEL: M37, L82, C61, C80

Keywords: Social media, influencers, reputation, single-agent dynamics

*I am indebted to Luís Cabral for exhaustive feedback on this project. I also thank Petra Moser, Paul Scott, Ambre Elsas-Nicolle, Matthew Mitchell, and many others.

†Kazimier Smith (kazimiersmith@gmail.com): Massachusetts Institute of Technology

1 Introduction

Social media influencers drive much of the modern digital economy. They produce the content to which the average American devotes two hours of their day (Statista, 2024a). They interact personally with fans and build loyal followings. They recommend products to these audiences, fueling the $220 billion social media marketing industry (Statista, 2024b). Large brands capitalize on the trust followers have in influencers to run more impactful advertising campaigns. Niche brands can use influencers to target a specific audience most likely to enjoy their products. For any firm, a product recommendation that goes viral can generate overwhelming sales. This ecosystem attracts broad interest: one in four members of Gen Z wants to become a social media influencer (Novacic, 2019).

Despite the economic relevance of the content influencers produce, the way influencers choose how much and what types of content to create remains poorly understood. Influencers can essentially produce two types of content. The first is *organic posts*, which are standard social media content that does not include a product recommendation. For example, a cooking-focused influencer films themselves making pizza dough and shares the recipe with their audience. The second type is *sponsored posts*. A sponsored post is like an organic post, but it includes a specific product recommendation. The cooking influencer films the pizza recipe and mentions that they prefer to use King Arthur Flour. King Arthur Flour pays the influencer for the post, and the payment depends on the size of the influencer's audience (their follower count).

Existing theoretical literature identifies the key dynamic tradeoff an influencer faces: sponsored content generates revenue but either reduces follower count or slows its growth. Nistor et al. (2024) assume that when an influencer starts endorsing products that are a poor fit for their audience, a certain fraction unfollows them. Mitchell (2021) models the length of the relationship between an influencer and a follower, but the same tradeoff arises: sponsored content reduces the length of the relationship. Determining an influencer's optimal content production strategy requires quantifying this tradeoff. If sponsored posts incur a large penalty, one might expect an investing and harvesting pattern where influencers make mostly organic posts until they reach a certain size, at which point they switch to sponsored posts to extract value from their audience. If the growth penalty for sponsored content is small, influencers might pursue a more balanced strategy to ensure they earn some income now and some in the future.

To fill this gap, I develop a structural model of an influencer's content production choice. I estimate the parameters of the model to quantify the tradeoff between organic and sponsored posts. The optimal policy gives the number of organic and sponsored posts an influencer should produce each week at a given follower count. For example, an influencer with 10,000 followers behaving optimally makes about two organic and 0.25 sponsored posts per week, while an influencer with 1,000,000 followers increases production to about 3.5 and 0.5 per week, respectively. The paper proceeds in four main steps.

First, I develop a model of influencer content production. The influencer solves a dynamic optimization problem, choosing the number of organic and sponsored posts to produce each week to maximize discounted lifetime income. Given their current follower count, the influencer expends costly effort to produce organic and sponsored posts. Sponsored posts incur an additional cost over organic posts which captures negotiating with the sponsoring brand, learning about the sponsored product, and other things. The influencer is paid for each sponsored post based on their follower count. The change in follower count from the current period to the next period depends directly on the number of organic and sponsored posts the influencer makes. It also depends on the average performance of the influencer's content (how many likes it receives). Follower growth includes a normally distributed random shock to account for factors like the platform's content distribution algorithm which the influencer cannot observe.

Second, to estimate the relationship between follower count and pay, I collect information on about 20,000 payments from brands to anonymous influencers. The data reveal a positive, linear relationship between follower count and pay for sponsored content. In line with anecdotal industry evidence, an influencer with 10,000 followers earns about $145 per sponsored post.

Third, I estimate the relationship between the number and types of posts an influencer makes and the change in their follower count. I collect the entire post history of 1,369 Instagram influencers in eight categories and a daily time series of each influencer's follower count. The post and follower count data reveal that organic and sponsored posts both generate positive growth, and the penalty for sponsored content is small. For an influencer with 100,000 followers who otherwise would have gained 1,000 followers, one extra organic post yields 39 additional followers while one extra sponsored post yields 18.

A key piece of this analysis is the effect of sponsorship on the attention (likes) the post receives. I measure this using a novel empirical strategy based on *cross-posts*, which are identical posts that an influencer uploads to multiple platforms. Effort is the same for both posts because they are identical. When the post is sponsored on one platform and not on the other, I can isolate the impact of sponsorship on performance. Without controlling for effort, a sponsored Instagram post receives about 30% fewer likes than its organic counterpart. After adjusting for effort, the effect decreases to about a 19% penalty for sponsorship. Much of the growth penalty therefore comes from the fact that sponsored posts receive fewer likes, but even this effect is quite small. While a sponsored post gets 19% fewer likes, that translates into only about six fewer followers.

Fourth, I estimate the parameters of the cost function for producing content. I take as given the relationship between follower count and pay and the effect of content choices on follower growth. I then apply the method of simulated moments to estimate the parameters of the cost function. The resulting model fits the data well. In both the data and the estimated optimal policy, influencers with more followers produce more organic and sponsored content. An influencer with 10,000 followers behaving optimally makes about two organic and 0.25 sponsored posts per week, while an influencer with 1,000,000 followers increases production to about 3.5 and 0.5 per week, respectively. Influencers with more followers make more sponsored content because brands pay them more and because the additional revenue outweighs the increasing marginal cost of another post. They make more organic posts because doing so generates a larger raw increase in follower count than it would for a smaller influencer. That increase translates to larger future payments, so organic content is more attractive for larger influencers. On net, influencers increase and then slightly decrease their share of sponsored content as they grow. Initially, the extra pay from sponsored content makes it more attractive, but at a certain size, the growth benefit from organic content outweighs pay, so the fraction of sponsored posts begins to decline.

With the model and estimated parameters in hand, I analyze key features of and possible changes to the influencer economy. I simulate a viral post in the model to test the role of luck in an influencer's career. While viral posts provide a short-term boost, influencers grow by exerting sustained effort over time. I test the literature's theoretical prediction that influencer behavior includes periods of "investing", when the influencer focuses on growth rather than revenue, and of "harvesting", when the influencer makes sponsored content to extract value from their audience (Nistor et al., 2024). My model predicts an increase in the fraction of sponsored content because influencers with more followers are paid more. Beyond about 100,000 followers, however, the extra growth from and lower cost of organic content become so attractive that influencers stop increasing their fraction of sponsored content. Observed content production choices are largely driven by short-term costs and benefits rather than dynamic incentives. Finally, I run counterfactual simulations. For example, increasing the parameter governing the cost of a sponsored post by about 20% (e.g.

through disclosure rules that make sponsored content more time-consuming) slightly reduces both organic and sponsored content production. Regulations that affect the return to sponsored content will likely have little impact on organic content, alleviating concerns that they might cause influencers to substantially reduce their content production.

2 Literature

I contribute to the literature on the economics of influencers by empirically testing a key mechanism in existing theoretical models. In these models, sponsored content incurs a dynamic or reputation cost, since it generates revenue but either reduces follower count or slows its growth. Nistor et al. (2024) assume that when an influencer starts endorsing products that are a poor fit for their audience, a certain fraction unfollows them. Mitchell (2021) models the length of the relationship between an influencer and a follower, but the same tradeoff arises: sponsored content reduces the length of the relationship. I use my post and follower count data to test the empirical validity of this assumption. Contrary to expectations, both organic and sponsored content have a positive effect on follower growth, and the magnitudes of the effects are almost identical. This result could be unique to Instagram, since Cheng and Zhang (2022) find a negative effect of sponsored YouTube videos on subscriber count. YouTube is a very different platform from Instagram, and Hughes et al. (2019) find that consumer reactions to sponsored content differ across platforms. If consumers view Instagram as a source of product recommendations and YouTube as an entertainment platform, they may be more amenable to sponsored content on Instagram. Advertising can also be useful for consumers, for example, through the signaling effect for experience goods theorized in Nelson (1974). Sahni and Nair (2020) find this effect to be large and positive for restaurants. Influencers in my sample often advertise experience goods: beauty influencers use specific brands of makeup, tech influencers recommend particular PC components, and food influencers favor one olive oil over another. If brands signal their quality by paying influencers to advertise, sponsored content could have a positive effect on consumers.

Behavioral literature empirically demonstrates several variables that mediate negative effects of sponsored content, such as explicit disclosure (Giuffredi-Kähr et al., 2022), the number of other people the influencer follows (Valsesia et al., 2020), and use of high-arousal language (Cascio Rizzo et al., 2024). Many factors simultaneously influence post performance, so pinning down the true effect of sponsorship is empirically challenging. Bairathi and Lambrecht (2023) use the Federal Trade Commission's warning to influencers about disclosure as an instrument to identify the effect of sponsorship and find a negative effect. I offer another empirical strategy leveraging *cross-posts* uploaded to both Instagram and TikTok. I show that sponsorship has a negative impact on follower engagement, but controlling for post quality using a counterfactual non-sponsored post reduces the magnitude of the effect. The studies above show varied impacts of sponsorship depending on the nature of the content, and my result indicates that influencers can mitigate sellout effects by making higher quality sponsored posts.

I also contribute to several parts of the literature on influencer marketing. The payment data I collect allow me to test the predicted relationships between pay and follower count and between pay *per follower* and follower count. Wies et al. (2023) hypothesize and empirically confirm an inverted U relationship between follower count and engagement with sponsored content. This confirms industry wisdom that the most valuable influencers on a per-follower basis are those in the middle of the follower count distribution. Brands running influencer marketing campaigns care about reaching many people but also about converting them to purchases, so they should pay more per follower to mid-sized influencers than to very large influencers.

Instead, I find a consistent negative relationship between pay per follower and follower count (Figure A.2). Precisely explaining this pattern requires further study, but it could reflect small influencers' unwillingness to work for low pay. If producing a sponsored post has some fixed cost independent of follower count, then brands must pay more per follower to small influencers to make their offers worthwhile.

Tian et al. (2024) predict an S-shaped relationship between impressions on a sponsored post and follower count, although the shape varies depending on the specific marketing campaign. If firms set pay by equating it to the marginal benefit (in terms of impressions) of another follower, then the relationship between pay and follower count should be similar to the relationship between impressions and follower count. I find a positive and approximately linear relationship (Figure A.1), so brands either are not aware of or do not internalize the declining marginal benefit of an additional follower. While smaller influencers may see higher engagement on their content, brands seem to care mostly about reach (i.e. follower count). If they took engagement into account, the pay vs. follower count curve would likely exhibit nonlinearities.

I use the pay and follower growth facts I establish to inform a dynamic model of influencer content production similar to the model of authenticity choice in Nistor et al. (2024). The influencer's key choice in their model is whether to accept sponsorship offers that are a poor fit for their audience. Accepting poor-fit offers generates more revenue but causes some followers to abandon the influencer. I focus on the distinction between sponsored and organic posts rather than the fit of sponsored content. This allows me to test whether the follower growth penalty or other costs primarily explain observed content production. Modeling organic and sponsored posts also adds a degree of flexibility since influencers can substitute between the two. Authenticity arises as either the total number or fraction of sponsored posts, and I can describe the choices from which a given level of authenticity derives.

I implicitly assume, as do existing models, that making more sponsored posts requires accepting poor-fit offers that alienate followers. Since making more sponsored posts decreases their average fit in my model, the influencer covers a broader range of topics as they grow. Gong (2021) establishes the same fact as a solution to the cold-start problem: influencers initially make niche content to attract users with specific tastes. As the influencer develops their reputation, they broaden their content to attract more users. They also produce more sponsored content because, as in my paper and other models, their larger follower base makes it more lucrative. I do not explicitly model followers or the fit between content and audience. Instead, they appear in a reduced form way via the cost function for sponsored content. The cost of a sponsored post increases as the influencer makes more of them because the influencer starts with the highest fit sponsorship offers and then accepts worse ones. Notably, Leung et al. (2022) find an inverted U relationship between follower-brand fit and the effectiveness of influencer marketing campaigns. Future work could develop a dynamic model that captures this fact.

In existing literature, the dynamic cost of sponsored content generates two distinct behaviors. First, an influencer spends time growing their audience by prioritizing organic content ("investing"). Once their audience is sufficiently valuable, they "harvest" by making sponsored content for brands in exchange for money. Nistor et al. (2024) and Mitchell (2021) both show this pattern in different ways. The influencer is willing to spend time investing because it generates higher future advertising revenues. Forgoing payment today can maximize total lifetime utility. My data show that this tradeoff is empirically small, so while it appears in my model, it is not the main force disincentivizing sponsored posts. Instead, the marginal benefit of another sponsored post increases with follower count more quickly than the marginal cost, so influencers make more sponsored content as their audiences grow.

Viral posts appear in my model because follower growth is subject to a random shock each period,

and these shocks are occasionally large and positive. While literature beginning with Berger and Milkman (2012) identifies factors that influence virality, influencers' post histories indicate that "going viral" is rare and unpredictable. Influencers instead try to post high-quality content regularly and hope that it will occasionally perform extremely well. When influencers do go viral in my model, they grow quickly for a short period of time before returning to their previous trend.

Only a few existing papers structurally estimate the parameters of an influencer's problem. Tang et al. (2012) model and estimate a YouTuber's content production decision. They include revenue sharing but do not explicitly model sponsored vs. organic content. Li (2023) studies the disclosure decisions of Twitch streamers and, in counterfactual analyses, examines the impacts of regulating disclosure. Instagram is a very different platform from YouTube and Twitch, so studying Instagram influencers is valuable in its own right. I also collect novel data on payments from brands to influencers so I can more accurately estimate how pay impacts influencer utility. Adding this relationship to the model means I can identify unknown cost function parameters. Modeling explicit costs of both types of content opens up new counterfactual scenarios to analysis.

3 Model

I model an influencer's content choice as a dynamic optimization problem in discrete time. The influencer is paid for sponsored posts, but producing organic and sponsored posts is costly. The influencer chooses the number of organic and sponsored posts to make each period to maximize their discounted lifetime utility. I define a period to be one week because in my data, influencers do not post every day; aggregating to the weekly level provides more variation in the number of posts. I allow fractional numbers of posts, so an influencer can make 3.6 organic posts. In the data I often observe influencers playing a "mixed strategy" like making a sponsored post every other week; I model this as making 0.5 sponsored posts in one week.

Concretely, if the influencer has f_t followers at the beginning of period t and makes o_t organic posts and s_t sponsored posts, their utility is

$$\alpha(f_t)s_t - c_e(o_t, s_t) - c_m(s_t) \tag{3.1}$$

where

- $\alpha(f_t)$ is the payment the influencer receives for making one sponsored post given follower count f_t,
- $c_e(o_t, s_t)$ is an "effort cost" of producing o_t organic posts and s_t sponsored posts, and
- $c_m(s_t)$ is an additional production cost that only sponsored posts incur.

I assume a log-linear relationship between pay for a sponsored post and follower count:

$$\log_{10} \alpha(f_t) = [\pi_0 + \pi_1 \log_{10}(f_t)] \tag{3.2}$$

for some coefficients π_0 and π_1. While brands might account for other factors when deciding how much to pay influencers, the simple relationship above fits the data I observe very well. It captures the standard industry fact that influencers with more followers are paid more.

The second term c_e is the *effort cost* of content; it captures the time and energy the influencer puts into making organic and sponsored posts. I assume that for some parameters θ_e and η,

$$c_e(o_t, s_t) = \theta_e(o_t + s_t)^\eta$$

In terms of pure effort, organic and sponsored content have the same cost: I assume setup, planning, filming, etc. takes the same amount of time for both types of post. The parameter η makes the cost function convex, so the marginal cost of one or two posts is low, but additional posts quickly become more costly because ideas are scarce. The influencer has a couple of good post ideas on which they can start working immediately. Making more posts requires thinking of new ideas which are increasingly difficult to generate.

In my data, the largest influencers make 4-5 posts per week on average, but fewer than one of those is sponsored. If utility depended only on the two components α and c_e described above, rationalizing observed behavior would require a large negative effect of sponsored content on future follower count. I do not observe such an effect in the regression analysis in Section 5. Sponsored content must therefore have unique costs that do not apply to organic content. To capture these I introduce a *match cost* c_m. The total match cost of s_t sponsored posts follows an exponential function according to a parameter θ_m:

$$c_m(s_t) = e^{\theta_m s_t} - 1$$

I remain agnostic on the specific interpretation of c_m; I think of it as generally capturing costs of producing sponsored posts that do not apply to organic posts. For example, to make more sponsored posts, an influencer must accept sponsorship offers from brands that are increasingly far from their typical content. A vegan recipe influencer might receive a few offers from vegan ingredient brands, but producing more sponsored posts requires accepting a sponsorship offer from a meditation app company. The latter sponsored post is more costly for the influencer because (1) they must find a way to incorporate meditation into their usual content (vegan recipes) and (2) the influencer feels bad about "selling out" to a brand in which their audience has little interest. Alternatively, making more sponsored content might require the influencer to search for brands with which to partner, creating a time cost unique to sponsored content. Finally, regulations could impose additional costs on sponsored content. If sponsored posts have to follow a specific format different from the influencer's usual post, then the influencer will have to spend time conforming to the requirements. Ultimately c_m is the reduced form of a more complex matching model between influencers and brands. My goal is to describe how influencer behavior changes in response to changes in the costs of making content. Modeling the matching process explicitly is an interesting avenue for future work.

Note that the marginal effort cost,

$$\theta_e \eta (o_t + s_t)^{\eta - 1}$$

goes to zero as $o_t + s_t$ goes to zero, so it is nearly costless to produce a very small quantity of content. For example, an influencer can film a one-second video of themselves and upload it to Instagram nearly instantly. This ensures influencers always have some incentive to produce content.

On the other hand, the marginal match cost,

$$\theta_m e^{\theta_m s_t}$$

approaches θ_m as s_t goes to zero. Producing sponsored content always has a nonzero cost. The influencer must always spend some time adapting a sponsoring brand's product to their personal style, and any sponsored post requires communicating with the brand and filling out paperwork. While sponsored posts are inherently more costly than organic posts in my model, they are also the only way the influencer can earn money.

The flow utility specification (3.1) is novel partly because it models production of organic and sponsored

content separately rather than employing an aggregate authenticity measure. Existing literature allows the influencer to choose the fraction of sponsored content they make or their probability of accepting a sponsorship offer, and the influencer can costlessly adjust these metrics (Mitchell, 2021; Nistor et al., 2024). In such a model, equilibrium content choices are driven by the influencer's reputation, which erodes if the influencer lowers their authenticity too much. By focusing on reputation, these models do not allow short-term costs of producing content to drive influencer behavior.

But social media posts are economic objects that require resources to produce; they have increasing marginal costs because time and ideas become scarce. Modeling content production explicitly allows me to quantify how much long-term reputation concerns versus short-term production costs drive observed choices. I can also examine substitution between the two types of content and the effects of changes in model parameters on each type separately. If, for example, a policy change yields a decrease in the fraction of sponsored content *and* a reduction in total content, then the apparent increase in authenticity (the lower fraction of sponsored posts) could harm consumers. The influencer can also expend effort to earn higher pay while maintaining a constant level of authenticity: an influencer who increases both sponsored and organic content will see current period revenue go up with no accompanying decrease in authenticity since the fraction of sponsored content stays the same. The optimal number of posts is governed by the model parameters, which I estimate from data.

Given the utility specification, the influencer chooses a sequence $\{o_t, s_t\}_{t=0}^{\infty}$ to maximize discounted lifetime utility

$$\sum_{t=0}^{\infty} \beta^t \left[\alpha(f_t)s_t - c_e(o_t, s_t) - c_m(s_t)\right]$$

where β is the discount factor. I summarize the problem in a Bellman equation in which the state is the influencer's follower count f_t:

$$V(f_t) = \max_{o_t, s_t} \alpha(f_t)s_t - c_e(o_t, s_t) - c_m(s_t) + \beta \mathbb{E}V(f_{t+1}) \tag{3.3}$$

I assume $\beta = 0.99$. The influencer's problem is dynamic because content choice (o_t, s_t) affects the evolution of follower count from f_t to f_{t+1}. To model the transition I assume

$$\log f_{t+1} - \log f_t = \tau_o o_t + \tau_s s_t + \tau_\ell \ell_t + \epsilon_t \tag{3.4}$$

where ℓ_t is the number of likes the influencer's content receives in period t and $\epsilon_t \sim \mathcal{N}(0, \sigma_f^2)$. The error term captures randomness in follower count growth that is unobservable to the influencer, such as variation in the platform's content distribution algorithm.

Likes measure the amount of positive attention a post receives. Influencers with more followers will get more likes on their posts since the content is distributed to a wider audience. Conditional on follower count, however, sponsored posts might receive fewer likes if followers find them annoying. A post with more likes attracts more new followers because, for example, the platform's content distribution algorithm shows it to more users. I assume that in a given time period, an influencer's posts receive a baseline number of likes ℓ_o proportional to the influencer's follower count f_t. Sponsored posts incur a penalty p so that influencers who make more sponsored content receive fewer likes. In particular, the final number of likes the influencer's content receives in period t is given by

$$\ell := \ell_o - p\frac{s}{o+s} \tag{3.5}$$

The penalty operates on the *fraction* of sponsored content because the influencer can dilute the negative impact of a sponsored post by making more organic posts. One sponsored post among five high-quality organic posts likely incurs a smaller penalty than one sponsored post that is the only content the influencer makes that period.

In summary, the influencer chooses how many organic and sponsored posts to produce each period. These choices affect the growth in the influencer's follower count, so producing more sponsored content generates higher revenue immediately but might lower revenue in the future. My primary contribution is quantifying the influencer's content production decision in light of this tradeoff: for a given follower count f_t, what is the optimal number of organic posts o_t and sponsored posts s_t to produce. I assume influencers behave according to the model above and then estimate the model parameter values that best explain observed choices by real influencers. There are four key pieces of the model that I take to the data:

1. How pay for sponsored content depends on follower count
2. How follower count evolves over time given content production
3. How content performance (likes) depends on content choices
4. How costly it is to produce organic and sponsored content

In the next few sections, I describe how I estimate the parameters of each piece.

4 Estimating influencer pay

Anecdotal evidence suggests that influencers with more followers are paid more for sponsored content; since their content reaches a wider audience, brands can market their product to a larger pool of potential buyers (Geyser, 2026). In my model, the influencer's optimal strategy depends on how exactly pay increases with follower count, and there is limited data on the precise relationship. To address this I collect a novel dataset from an anonymous online platform where influencers submit reviews of their collaborations with brands.[1] Each review corresponds to a single collaboration. Influencers report the name of the brand, their follower count, the pay they received for the collaboration, how many posts they created for the brand, and other information. Site staff verify both the influencers and the collaborations. The platform's primary goal is to make it easier for influencers "to figure out what price to charge for their services" (FYPM, 2024); they can use the site to see how much similar influencers are paid for brand collaborations.

Figure 4.1 shows a sample review. The influencer who submitted the review collaborated with a brand called "Cat Person" in late 2023. The influencer has 100,000 TikTok followers and 20,000 Instagram followers, and the brand asked them to post a short-form video on both platforms. The brand paid the influencer $500 cash and gave them cat chews worth $60. The "review" section at the bottom is the influencer's free-form description of their experience working with the brand.

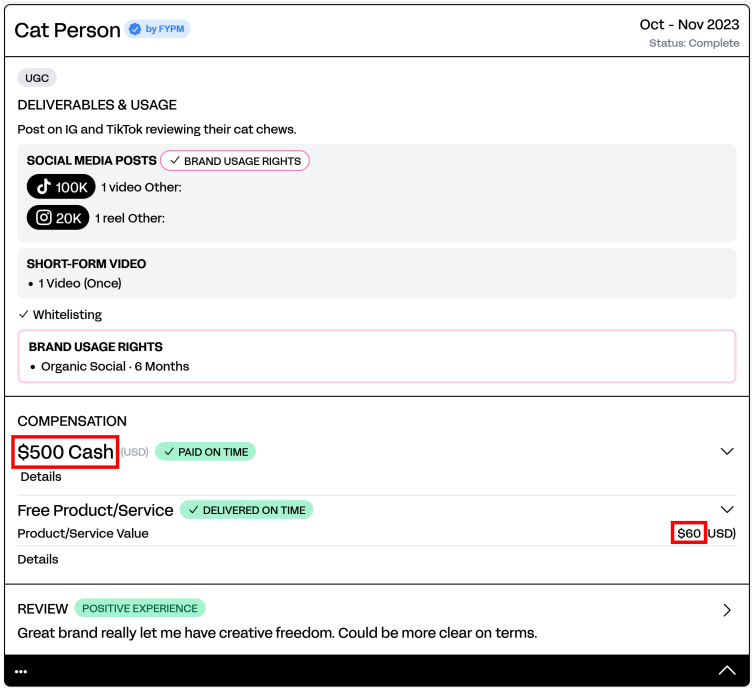
From these reviews, I estimate the relationship between follower count and pay. I average all the listed follower counts in the review,[2] so in the example above the influencer's aggregated follower count is

$$\frac{100,000 + 20,000}{2} = 60,000$$

[1] fypm.vip

[2] I obtain similar coefficient estimates when I use only Instagram followers, reported in Table A.1

Figure 4.1: Review of brand collaboration



I sum cash pay and the value of free products so total compensation is \$560.[3] Finally, I adjust for the fact that collaborations requiring multiple pieces of content tend to pay more (Figure A.4). The platform reports an aggregate number of "deliverables" for each review; it is three in the example above. My outcome variable is pay per deliverable, which is $\$560/3 = \$186.66$.

I collect 21,854 reviews and exclude those for which follower count, number of deliverables, or pay is zero. The site also includes reviews of *declined* collaborations, where a brand offered a sponsorship opportunity but the influencer turned it down. I exclude these since they do not describe the true relationship between follower count and pay for a sponsored post; influencers typically describe declined collaborations as paying too little to be worthwhile. The final sample has 15,047 reviews.

Figure 4.2: Main review data variables

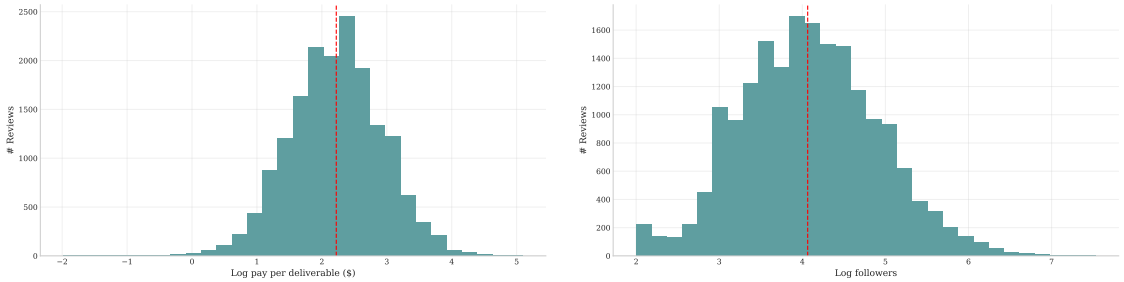


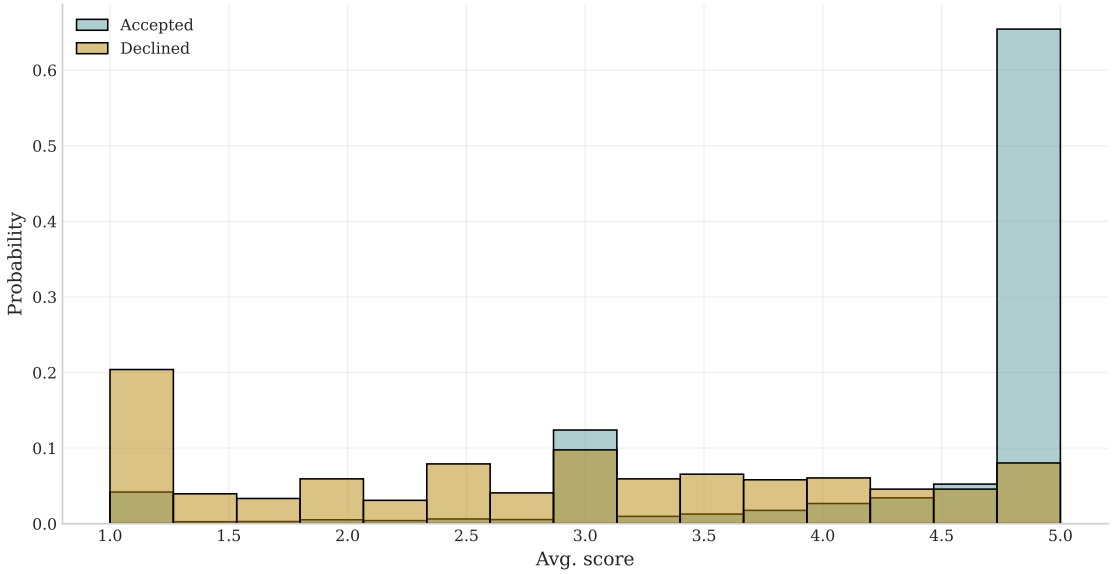
Figure 4.2 shows the distributions of the two main variables from the review data. The median accepted review comes from an influencer with about 10,000 followers who is paid about \$125 per deliverable.

[3]Reviews occasionally report pay in other currencies; I convert these to US dollars using the rates in Table A.2.

Sample selection. Influencers voluntarily contribute reviews to FYPM.vip, so the payment data could suffer from selection bias. If influencers only review positive experiences, pay might be inflated upwards. But the presence of 809 declined reviews means influencers also review *very bad* collaboration offers. Section A.1 describes these in more detail. If influencers are willing to post unacceptable offers on the platform, they are likely also willing to post low-paying and low-quality offers. Moreover, my estimates are broadly in line with industry estimates of $100 per 10,000 followers (Geyser, 2026). If anything, the industry numbers suggest that larger influencers are paid slightly more than my estimates imply. This means upward bias in the pay data is unlikely.

On the other hand, if influencers use the platform as a way to publicly shame brands that offer low pay, my estimates might be biased downwards. But influencers rate the majority of accepted reviews positively. The platform allows influencers to rate the brand collaboration on four dimensions (Figure A.3). I assign

Figure 4.3: Review scores



each review an average score by taking the mean of the star ratings in each category. Figure 4.3 shows the distribution of these average scores separately for accepted and declined reviews. The majority of accepted reviews receive close to five stars in all categories, so influencers seem to mostly review collaborations with which they are satisfied. If influencers were mostly reporting low-paying collaborations, reviews with fewer than five stars would likely be more common. Instead, it appears that influencers simply reject sponsorship offers that pay too little.

Results. I use the pay data to estimate the coefficients in Equation 3.2. Table 4.1 reports the estimates of π_0 (the intercept) and π_1 (the coefficient on log followers).

Since date fixed effects do not substantially increase explanatory power, I use $\pi_0 = 0.206$ and $\pi_1 = 0.489$ when estimating the remaining parts of the model. An influencer with 10,000 followers receives $145 per deliverable, while pay increases to $448 per deliverable for an influencer with 100,000 followers. The numbers align with industry ballpark numbers: Geyser (2026) reports $100 to $500 per post for "micro-influencers" with 10,000 to 100,000 followers. With the first key piece of the model estimated, I move to the effect of the influencer's content production choices on the evolution of their follower count.

Table 4.1: Dependence of pay on follower count

	Log pay per deliverable	Log pay per deliverable
Log followers	0.489***	0.482***
	(0.006)	(0.006)
Intercept	0.206***	1.878**
	(0.024)	(0.582)
Date FE	No	Yes
N	15,047	15,047
R2	0.327	0.337

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

5 Estimating the follower count transition function

Sponsored content is the primary way influencers earn money, and Section 4 establishes that pay increases with follower count. An influencer who cares about current and future income must therefore consider how their present choices affect how many followers they will have in the future. I quantify this effect using post and follower count histories for 1,369 influencers listed on feedspot.com. The site maintains pages like "Top 100 Lifestyle Influencers", and I collect these pages for the categories beauty, cycling, fitness, food, lifestyle, mom, tech, and travel. Each list includes influencers of varying sizes; the smallest influencer in my sample had 883 Instagram followers on January 1st, 2023, while the largest had nearly 52 million.

Next, I use a data collection platform to collect every Instagram post from every influencer in my sample.[4] After dropping duplicate posts (which likely arise from scraping issues), posts missing a username, and posts missing a date, I end up with 2,780,011 Instagram posts. For each of these I collect all text-based information about the post, including the post caption, hashtags, number of likes, post date, comments, any paid partnership labels, and other features. Figure 5.1 shows a sample post from the recipe creator "@dadaeats", which is representative of content from food influencers. The post describes a recipe and includes an edited video of the creator preparing the recipe, along with a set of hashtags that categorize the content.

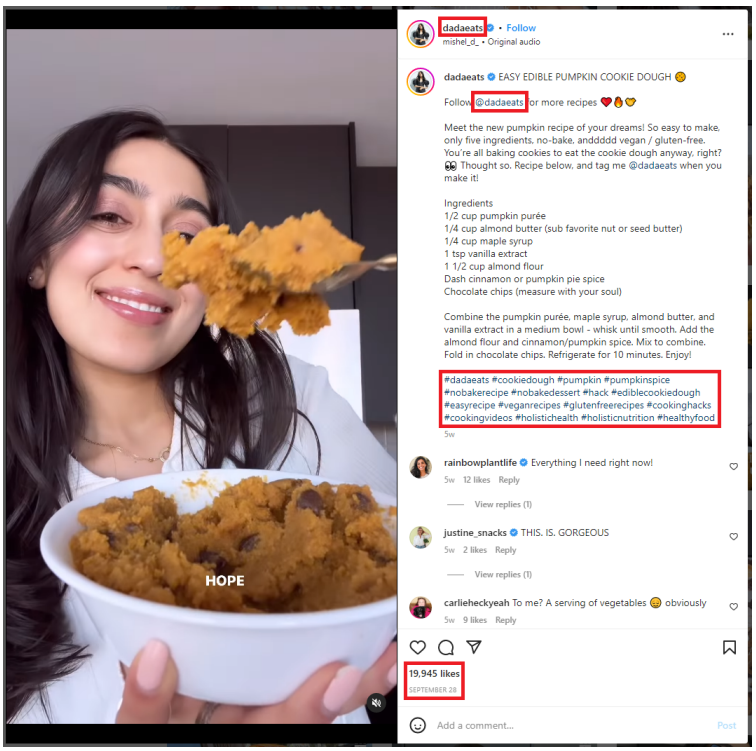
5.1 Classifying sponsored posts

The influencer faces a tradeoff between organic posts, which might generate higher future revenue by attracting new followers, and sponsored posts, which generate revenue immediately. To quantify this tradeoff empirically, I first classify every Instagram post in the sample as organic or sponsored. I analyze post text to classify sponsored posts in my data. I classify a post as sponsored if it meets at least one of the following conditions:

1. Uses the official platform disclosure tool (Paid partnership with...)
2. Contains explicit disclosure like #ad, #sponsored, or #lululemon_partner
3. Mentions one of 8,094 brands (a mention begins with the @ symbol)

[4] Brightdata through the Bright Initiative.

Figure 5.1: Example Instagram post

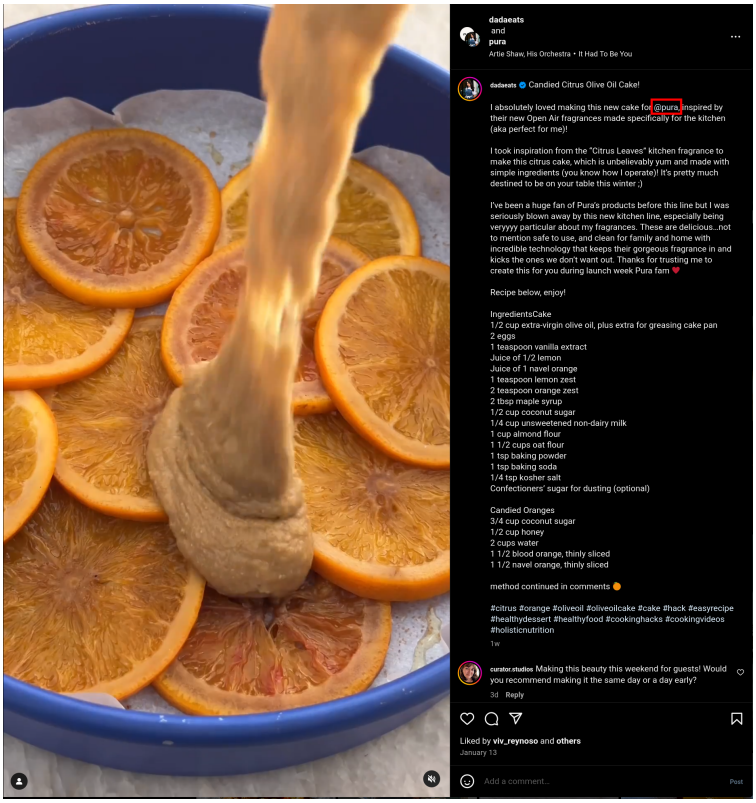


The first two points are clear evidence that the post was financially supported by a brand. However, inspecting posts in my sample revealed that influencers often indicate sponsorship only by mentioning the name of the brand in the post text. For example, Figure 5.2 shows a post sponsored by Pura, a home fragrance producer. The post lacks the paid partnership label but is clearly sponsored because it explicitly mentions the brand and because one of the brand's products inspired the recipe. I address this by collecting the names of all sponsoring companies in the review data described in Section 4. Since a brand's Instagram username might not match its official name, I create a list of several possible usernames for each brand: the name converted to lower case, the name with spaces removed, and the name with spaces replaced by underscores, dots, or dashes. I search the text of each post for a mention of any of these brand usernames and classify the post as sponsored if a mention appears. Appendix Section B.1 discusses alternative sponsorship classification methods.

Table 5.1: Organic vs sponsored Instagram posts

Variable	Organic	Sponsored	p-value
Followers	853,995	822,795	0.000
Likes	10,950	11,093	0.129
# Comments	155	211	0.000
# Words in post	52	86	0.000
Viral	0.063	0.1	0.000
N	2,532,763	247,248	NA

Figure 5.2: Sponsored post with brand mention



After classifying all posts, I find that 244,202 Instagram posts (8.78%) are sponsored. Table 5.1 reports the results of a two-sided Student's t-test comparing organic and sponsored posts on several dimensions. Sponsored posts have slightly longer text, possibly a mechanical effect from describing the sponsoring brand. Follower count at the time of posting is similar for organic and sponsored posts. The difference in average likes for both post types is statistically insignificant, but the t-test does not control for follower count. Having separated organic and sponsored posts in the data, I move to estimating the effects of each on the size of the influencer's audience.

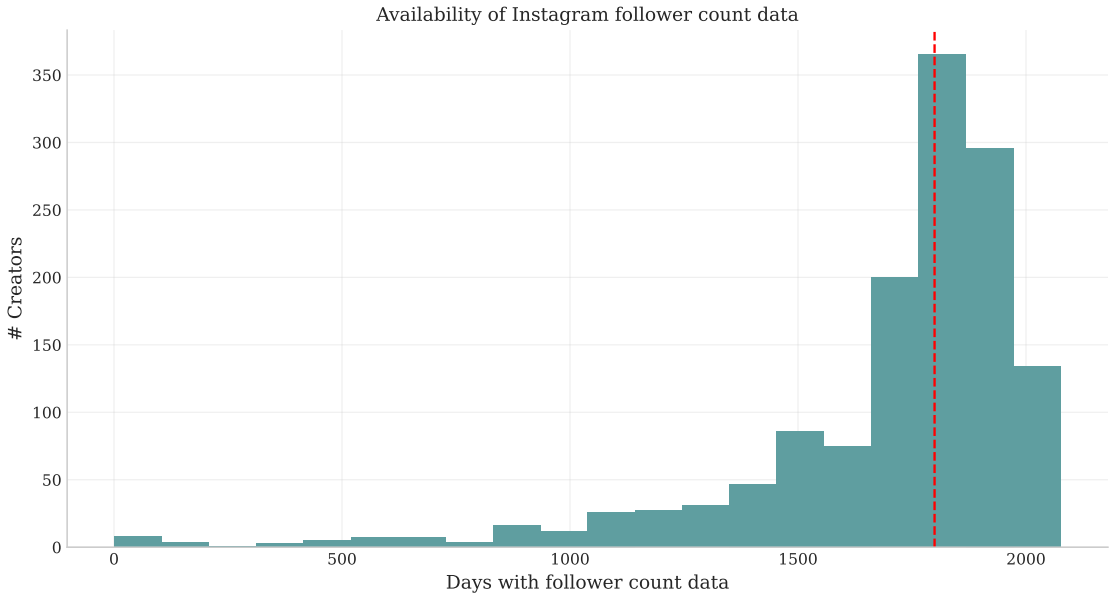
5.2 Estimating the parameters of the transition equation

In order to numerically solve the Bellman equation (3.3) I need to calculate the continuation value $\mathbb{E}V(f_{t+1})$. Next period's state f_{t+1} depends on the state f_t and the influencer's choice variables o_t and s_t through Equation 3.4. Define ϕ to be the density of $\log f_{t+1}$ conditional on f_t, o_t, s_t, σ_f, that is, ϕ is the density of a random variable with distribution $\mathcal{N}(\log f_t + \tau_o o_t + \tau_s s_t + \tau_\ell \ell_t, \sigma_f^2)$. Then I can calculate the continuation value by integrating with respect to ϕ as detailed in Section 7.1. The immediate empirical task is to estimate $\tau_o, \tau_s, \tau_\ell$, and σ_f.

To do this, I combine the post data described above with time series data on follower counts. I collect daily Instagram follower count histories from a social media analysis platform that tracks influencers over time.[5] The data provide a long and detailed view of follower growth for each influencer; as shown in Figure 5.3, I am able to collect nearly five years of daily data for the median influencer. I aggregate the data to

[5]Social Blade

Figure 5.3: Instagram follower count data availability



form a weekly panel of influencers and estimate the following specification:

$$\log f_{it+1} - \log f_{it} = \tau_o o_{it} + \tau_s s_{it} + \tau_\ell \ell_{it} + \epsilon_{it} \tag{5.1}$$

where

- f_{it+1} is influencer i's follower count in week $t+1$
- f_{it} is influencer i's follower count in week t
- o_{it} is the number of organic posts by influencer i in week t
- s_{it} is the number of sponsored posts by influencer i in week t
- ℓ_{it} is the average number of likes across all posts by influencer i in week t, or zero if the influencer did not post
- ϵ_{it} is a normally distributed influencer-specific follower count shock

I use the *change* in follower count as the outcome variable because of the unit root problem, which I detail in Section D.1.

Table 5.2 presents my estimates of the key transition function parameters. Organic posts, sponsored posts, and likes all have small but positive and statistically significant effects on follower growth. The mean of the regression residuals is $1.06 \cdot 10^{-18}$, validating my assumption that the random shock to follower growth is mean zero. For subsequent calculations I set σ_f to the standard deviation of the regression residuals, about 0.008. The most surprising result is the similarity between the estimates of τ_o and τ_s. Organic and sponsored content appear to have similar effects on future follower count. Although the coefficient on sponsored posts is smaller, the difference between the two coefficients implies very little difference in follower count growth. The result challenges the common perception that sponsored content leads to slower or even negative follower

Table 5.2: Empirical transition function

	Change log followers	Change log followers	Change log followers
Posted	0.00077***		
	(4.664e-05)		
# Organic posts		0.00025***	0.00019***
		(1e-05)	(1e-05)
# Sponsored posts		0.00016***	0.00010***
		(2e-05)	(2e-05)
Posted * Log likes			0.00028***
			(2e-05)
N	301,392	301,392	301,392
Creator FE	Yes	Yes	Yes
Date FE	Yes	Yes	Yes
Category FE	Yes	Yes	Yes
R2	0.000917	0.0036	0.00451

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

growth, and it is robust to a variety of alternative models of the follower count transition function, which I document in Section D.2.

The fact that τ_o and τ_s are positive and nearly identical in magnitude implies that the dynamic, reputational cost of sponsored content is low. If τ_s were large and negative, influencers would have a strong incentive to produce organic content to grow their audience. They would only begin making sponsored content when the reward was sufficiently large, leading to the invest-then-harvest mechanism explored in the theoretical literature (Mitchell, 2021). My empirical results suggest this mechanism is unlikely. The data, however, show that influencers make more than twice as many organic posts as sponsored posts (Figure 7.1). Since organic and sponsored posts have similar dynamic effects, the difference must be driven by current-period differences in their production costs, which I estimate in Section 7. The absence of a large dynamic penalty for sponsored content could be platform dependent. Consumers use Instagram partly to discover new products, so they may see sponsored content as no less valuable than organic content (which entertains them). In contrast, a platform like YouTube could be seen as pure entertainment, in which case users might punish content creators for advertising too much (Cheng and Zhang, 2022).

When estimating the parameters of the transition function, I use the observed number of likes each post in my data receives. Likes, however, are not a choice variable for the influencer; they accrue based on the number of organic and sponsored posts the influencer makes, as I describe in Section 3. In the next section, I quantify the effect of post type on likes.

6 Effect of sponsorship on content performance

The amount of attention an influencer's content receives, measured by likes, affects the growth of the influencer's audience. This means that solving the influencer's Bellman equation (3.3) requires quantifying how the influencer's content choices affect the number of likes they receive. While both organic and sponsored

posts can receive likes, sponsored posts might incur a penalty relative to organic posts, as described in Section 3. To quantify this penalty I need to estimate two objects:

1. The baseline number of likes ℓ_o a post receives for an influencer with a given follower count
2. The penalty p for sponsored posts

6.1 Estimating baseline likes

I calculate ℓ_o by regressing the number of likes a post received on the author's follower count using the 914,810 organic posts in my Instagram data. Table 6.1 shows the result: a strong positive relationship

Table 6.1: Effect of follower count on likes, organic posts only

	Log Instagram likes
Log followers	0.664***
	(0.000)
N	914,810
R2	0.979

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

between follower count and likes. Since the regression has no controls, the coefficient estimate is the mean number of likes on a post from an influencer with a given number of followers. The high R^2 indicates that follower count explains most of the variation in likes. Influencers receive less than one like per follower, possibly because not all followers engage with every post. When estimating the parameters of the cost function in Section 7, I fix $\ell_o = 0.664$.

6.2 Estimating the sponsorship penalty

Estimating the sponsorship penalty p is more difficult because sponsored posts might get fewer likes for two reasons. First, followers dislike advertising. Second, influencers might put less effort into sponsored posts, in which case the posts perform poorly because they are low quality, not because they are sponsored.

To disentangle these two factors, I use the fact that influencers sometimes upload identical content to multiple platforms. In particular, some of the Instagram posts in my data also appear on TikTok. I call these pairs of identical posts *cross-posts*. Figure 6.1 gives an example. The post on the left (Instagram) is sponsored since it includes a "paid partnership" label. The post on the right (TikTok) includes no clear sponsorship disclosure. In these cases, I use the TikTok post as a counterfactual organic version of the post; it proxies for the number of likes the post would have received had it not been sponsored. This strategy assumes that viewers treat the Instagram version as sponsored and, because the TikTok version has no sponsorship disclosure, viewers treat it as organic, and it receives likes accordingly.

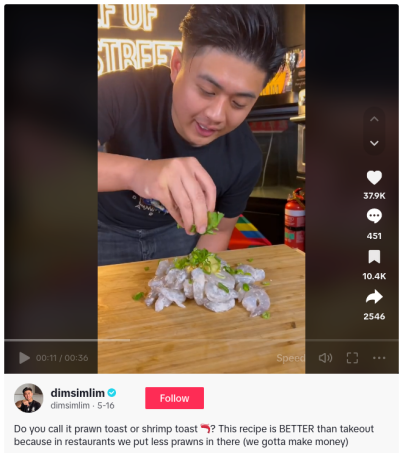
To construct pairs of cross-posts, I first manually match each Instagram account in my data to the corresponding TikTok account by searching Google, checking the influencer's Instagram profile, and checking the influencer's personal website. I find a TikTok account for about 80% of the influencers in my sample. From these TikTok accounts, I collect 136,453 individual posts.[6] Table 6.2 compares several post characteristics on

[6] Brightdata through the Bright Initiative.

Figure 6.1: Example cross-post



Sponsored



Not sponsored

the two platforms. Average follower count and average likes are higher on TikTok, likely because of selection

Table 6.2: Comparing Instagram and TikTok posts

Variable	Instagram	TikTok	p-value
Followers	849,977	1,080,454	0.000
Sponsored	0.089	0.091	0.015
Likes	10,962	28,493	0.000
# Comments	160	133	0.000
# Words in post	55	25	0.000
Viral	0.067	0.44	0.000
N	2,780,011	136,453	NA

bias. For example, if maintaining a presence on two platforms requires more resources, influencers might not start using TikTok until they have a sufficiently large audience on Instagram. Similar selection could explain the higher average likes on TikTok posts.

Matching procedure. To construct pairs of cross-posts, I match Instagram and TikTok posts in my sample using the following procedure. First, I restrict the sample to video posts since TikTok is primarily a short-form video platform.[7] Next, I require both posts in the matched pair to be from the same influencer. I then iterate over all pairs of posts from the same influencer. For each possible pair, I use a standard dynamic programming algorithm to find the longest sequence of words present in the text of both posts. If this sequence is at least seven words long, I classify the pair as a match. Inspecting these pairs of matched posts suggests that pairs with at least seven consecutive words overlapping are almost always an identical video with identical text. There are a few cases where a single Instagram post matches to multiple TikTok posts; these appear to be series of posts which all partner with a specific brand and include boilerplate text.

[7] TikTok added support for "carousel" photo posts a bit before my data analysis; I assume these are not a significant portion of content on the platform.

To remove these, I classify a pair as matched if both members of the pair have only one match on the other platform.

Matching cross-posts based on identical sequences of consecutive words is a somewhat conservative approach, and it likely excludes some pairs of posts that have identical videos but slightly different text. For my empirical strategy, it is better to be overly conservative. Having fewer matched pairs only reduces the sample size, while using an incorrectly matched post as a control would invalidate the empirical results.

Empirical strategy. To estimate the effect of sponsorship on likes, I use a sample of 9,697 Instagram posts that I am able to match to identical non-sponsored TikTok posts. As a baseline, I ignore the TikTok posts, and I regress the number of likes the Instagram post received on a dummy variable indicating whether or not the post is sponsored. I control for the influencer's follower count since a larger follower count means more people will see and possibly like the post. I include influencer and time fixed effects since sponsorship might affect likes differently across creators and over time. This baseline regression ignores the effect of post quality, so a negative coefficient on the sponsorship variable might come from audience distaste for sponsored posts, or it might come from lower content quality.

To control for post quality, I run a second regression which includes the variables above and adds the number of likes the post received on TikTok. Since the sample consists of matched pairs of posts, with identical videos, where the TikTok post is not sponsored, the TikTok post acts as a counterfactual non-sponsored version of the Instagram post. The number of TikTok likes, therefore, proxies for the quality of the post, allowing me to isolate the effect of sponsorship on likes.

Results. Table 6.3 shows the estimated coefficients from the two regressions described above. In the

Table 6.3: Effect of sponsorship on likes, controlling for quality

	Log Instagram likes	Log Instagram likes
Log Instagram followers	0.657***	0.489***
	(0.001)	(0.003)
Sponsored (Instagram)	-0.155***	-0.090***
	(0.027)	(0.022)
Log TikTok likes		0.321***
		(0.005)
Creator FE	Yes	Yes
Date FE	Yes	Yes
N	9,697	9,697
R2	0.970	0.980

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

baseline regression, sponsorship is associated with an approximately 30% decrease in the number of likes a post receives. In the second regression, the coefficient remains negative but shrinks in magnitude to imply a 19% reduction in likes. In the baseline regression, therefore, slightly less than half the negative effect of sponsorship is actually due to lower quality of sponsored posts. The remainder of the negative effect can be interpreted as the audience reacting negatively to sponsorship itself.

I have now estimated the two key pieces of the relationship between sponsorship and post performance (measured by likes). In the following sections, I take these estimates, $\ell_o = 0.664$ and $p = 0.090$, as given, and I estimate the remaining parameters of the model to fit observed data.

7 Estimating the cost function

I have estimated the influencer's payment function (how much influencers are paid for sponsored posts) and the transition function (how content choices affect future follower count). To calculate an influencer's optimal policy, the remaining task is quantifying the cost functions c_e and c_m. The parameters of these functions govern how the current-period production cost of content changes with the amount of content produced. In particular, θ_e affects the effort cost of producing any type of post, and θ_m affects the costs that only apply to producing sponsored posts. For the effort cost function c_e, I fix $\eta = 6$ so that the marginal cost of an additional post is convex, since I cannot separately identify θ_e and η using the data I have.

My estimation strategy has three main steps. First, I fix a guess of θ_e and θ_m. Second, I calculate the influencer's optimal policy, given the chosen parameter values. Third, I check how well the calculated optimal policy matches the observed data, and then I update my guess of the parameters. I iterate on this procedure until the parameter values change very little between iterations. Below, I describe how I calculate the value function, and then I detail my estimation procedure.

7.1 Value function calculation

The state variable in the Bellman equation (3.3) is the influencer's follower count. Solving the Bellman equation requires calculating the value function at every possible value of the state variable; since follower count can have millions of possible values, this is computationally intractable. The standard solution to this problem is to group follower counts into a smaller number of bins (Judd, 1998). However, in my data, follower counts change very little from week to week. A discretized value function would require a large and computationally burdensome number of grid points to properly approximate the follower count transition process. Instead, I assume the value function is continuous in follower count. Since more followers imply higher pay, the value function should be monotonically increasing, so I approximate it with a combination of cubic Hermite splines.[8] This method means I only need to calculate the parameters of the approximation, which greatly reduces the computational cost of solving the Bellman equation (Judd, 1998).

I allow the state variable (log follower count) to range over the interval $(2, 9)$, which interval contains all the observed follower counts in my data and extends beyond the maximum observed follower count. Limiting the grid to span only observed follower counts would force the value function to decline artificially as it approached the right endpoint of the grid. I interpolate the value function with $K = 40$ grid points $g_1, \ldots, g_K$ evenly spaced on the state space.

The algorithm to calculate the value function is a version of modified policy iteration (Judd, 1998). Let C be the polynomial approximation to the value function. I initialize it to be zero everywhere. I initialize the policy function for organic posts as the linear interpolation between one and five. I initialize the policy function for sponsored posts as the linear interpolation between zero and one. Let $o^*(f)$ be the optimal number of organic posts given the current optimal policy and f followers, and let $s^*(f)$ be the optimal number of sponsored posts. The algorithm proceeds as follows:

[8] I avoid linearly interpolating the value function because I calculate the expected value function with numerical integration. A non-differentiable value function slows down this calculation.

1. Let $\mathbf{V} := (C(g_1), \ldots, C(g_K))$ (the current value function evaluated at each grid point).

2. For $k = 1, \ldots, K$, calculate W_k as

$$W_k := \alpha(g_k)s^*(g_k) - c_e(o^*(g_k), s^*(g_k)) - c_m(s^*(g_k)) + \beta \int_2^9 C(x)\phi(x; g_k, o^*(g_k), s^*(g_k), \sigma_f)\, dx$$

   I calculate the integral with numerical quadrature.

3. Update C to be the piecewise cubic Hermite interpolating polynomial for $g_1, \ldots, g_K$ and $W_1, \ldots, W_K$.

4. Perform M iterations of Step 2 and Step 3. Afterwards, each W_k represents the value of following the optimal policy (o^*, s^*) for M periods given g_k followers in the first period.

5. For $k = 1, \ldots, K$, let

$$(o_k^{**}, s_k^{**}) := \underset{o,s}{\operatorname{argmax}}\ \alpha(g_k)s - c_e(o, s) - c_m(s) + \beta \int_2^9 C(x)\phi(x; g_k, o, s, \sigma_f)\, dx$$

6. Update the optimal policy: update $o^*(f)$ to be the piecewise cubic Hermite interpolating polynomial for $g_1, \ldots, g_K$ and $o_1^{**}, \ldots, o_K^{**}$, and update $s^*(f)$ to be the piecewise cubic Hermite interpolating polynomial for $g_1, \ldots, g_K$ and $s_1^{**}, \ldots, s_K^{**}$.

7. Let $\mathbf{V}' := (C(g_1), \ldots, C(g_K))$ (the new value function evaluated at each grid point). Let $\overline{\mathbf{V}}'$ be the maximum of the elements of $\mathbf{V}'$. Define

$$\iota := \frac{\|\mathbf{V} - \mathbf{V}'\|}{\overline{\mathbf{V}}'}$$

8. If $\iota \leq 10^{-5}$, stop, and use C as the final value function and (o^*, s^*) as the final policy function. Otherwise, return to Step 1.

This algorithm allows me to calculate the influencer's value function given particular values of model parameters. In the next section, I describe my method to estimate θ_e and θ_m, which requires repeatedly calculating the value function.

7.2 Estimation methodology

I use the method of simulated moments to estimate the cost parameters θ_e and θ_m.[9] The general approach is to fix a guess of the parameters, simulate the posting behavior of many influencers, and then update the parameter guess based on how closely the simulated numbers of posts match observed behavior from the data.

7.2.1 Moment conditions

The final step of this approach requires moment conditions that measure the difference between the simulations and the data. I construct these using the average numbers of organic and sponsored posts created by simulated influencers with different follower counts. Let o_{it} and s_{it} be the numbers of organic and sponsored posts, respectively, that individual i makes in period t in the data. Let $\overline{o}_b^*$ and $\overline{s}_b^*$ be the average numbers of

[9] I thank Li et al. (2016), whose paper provided me an excellent guide to understanding and implementing the methodology.

organic and sponsored posts from *simulated* influencers with follower counts in each of B bins, $b = 1, \ldots, B$. There are $2B$ moment conditions in total; for each bin they are

$$\begin{aligned}\mathbb{E}\left[o_{it} - \overline{o}_b^* \;\middle|\; \text{individual } i \text{ in bin } b \text{ at time } t\right] &= 0 \\ \mathbb{E}\left[s_{it} - \overline{s}_b^* \;\middle|\; \text{individual } i \text{ in bin } b \text{ at time } t\right] &= 0\end{aligned}$$

To convert these to a form suitable for estimation, define

$$g^b(o_{it}, s_{it}, \theta_e, \theta_m) := \begin{bmatrix}(o_{it} - \overline{o}_b^*)\mathbb{1}(\text{individual } i \text{ in bin } b \text{ at time } t) \\ (s_{it} - \overline{s}_b^*)\mathbb{1}(\text{individual } i \text{ in bin } b \text{ at time } t)\end{bmatrix} \tag{7.1}$$

Let $g(o_{it}, s_{it}, \theta_e, \theta_m)$ be the $2B \times 1$ column vector formed by stacking $g^1, \ldots, g^B$. The *unconditional* moment conditions are then given by

$$\mathbb{E}\left[g(o_{it}, s_{it}, \theta_e, \theta_m)\right] = 0$$

When taking these conditions to the data, I calculate sample analogs by summing over individuals $i = 1, \ldots, I$ and time periods $t = 1, \ldots, T$. For example, for bin b and parameters (θ_e, θ_m), the sample analog of $g^b(o_{it}, s_{it}, \theta_e, \theta_m)$ is

$$g_{IT}^b(\theta_e, \theta_m) = \begin{bmatrix}\frac{1}{IT}\sum_{t=1}^T \sum_{i=1}^I o_{it}\mathbb{1}(i \text{ in } b \text{ at } t) - \frac{\overline{o}_b^*}{IT}\sum_{t=1}^T \sum_{i=1}^I \mathbb{1}(i \text{ in } b \text{ at } t) \\ \frac{1}{IT}\sum_{t=1}^T \sum_{i=1}^I s_{it}\mathbb{1}(i \text{ in } b \text{ at } t) - \frac{\overline{s}_b^*}{IT}\sum_{t=1}^T \sum_{i=1}^I \mathbb{1}(i \text{ in } b \text{ at } t)\end{bmatrix}$$

Let $g_{IT}(\theta_e, \theta_m)$ be the $2B \times 1$ column vector formed by stacking $g_{IT}^1, \ldots, g_{IT}^B$.

7.2.2 Algorithm

Having defined the moment conditions, I use the following algorithm to estimate the values of θ_e and θ_m:

1. Set $N = 1000$ to be the number of simulations and $T = 200$ the number of periods per simulation. Draw starting follower counts f_n^0 for $n = 1, \ldots, N$ from the empirical distribution of follower counts on June 30, 2019. For $n = 1, \ldots, N$ and $t = 1, \ldots, T$, draw follower transition shocks $\epsilon_{nt} \sim \mathcal{N}(0, \sigma_f^2)$.

2. Fix a guess of (θ_e, θ_m). I start with $(0.0025, 9.5)$.

3. Use the algorithm in Section 7.1 to calculate the optimal policy. Let $o^*(f; \theta_e, \theta_m)$, $s^*(f; \theta_e, \theta_m)$, and $\ell^*(f; \theta_e, \theta_m)$, denote the optimal numbers of organic posts, sponsored posts, and likes, respectively, for an influencer with f followers, given cost parameters θ_e and θ_m.

4. Simulate the careers and content choices of each influencer n for T weeks as follows:

   (a) Set $f_{nt} = f_n^0$

   (b) Use the optimal policy to find $o^*(f_{nt}; \theta_e, \theta_m)$ and $s^*(f_{nt}; \theta_e, \theta_m)$.

   (c) Set $\log f_{nt+1} = \log f_{nt} + \tau_o o^*(f_{nt}; \theta_e, \theta_m) + \tau_s s^*(f_{nt}; \theta_e, \theta_m) + \tau_\ell \ell^*(f_{nt}; \theta_e, \theta_m) + \epsilon_{nt}$

   (d) Repeat steps (b) and (c) to generate $f_{n1}, \ldots, f_{nT}$.

5. Generate $B = 14$ equally sized follower count bins. For each bin b, calculate $\overline{o}_b^*$ and $\overline{s}_b^*$.

6. Repeat Steps 2–5 to find the first-stage estimates $(\widehat{\theta}_e^1, \widehat{\theta}_m^1)$ that minimize

$$g_{IT}(\theta_e, \theta_m)^\top g_{IT}(\theta_e, \theta_m)$$

7. Calculate the weighting matrix

$$\widehat{W} = \widehat{S}(\widehat{\theta}_e^1, \widehat{\theta}_m^1)^{-1}$$

where $\widehat{S}$ is an estimate of the covariance matrix of the influencer-level moment contributions (Appendix Section E).

8. Repeat Steps 2–5 to find $(\widehat{\theta}_e, \widehat{\theta}_m)$ that minimize

$$g_{IT}(\theta_e, \theta_m)^\top \widehat{W} g_{IT}(\theta_e, \theta_m)$$

Finally, I calculate the standard errors of my estimates of $\widehat{\theta}_e$ and $\widehat{\theta}_m$ using the weighting matrix $\widehat{W}$. The algorithm is described in Appendix Section E.

7.3 Results

Table 7.1 summarizes my estimates of the parameters of the model described in Section 3.

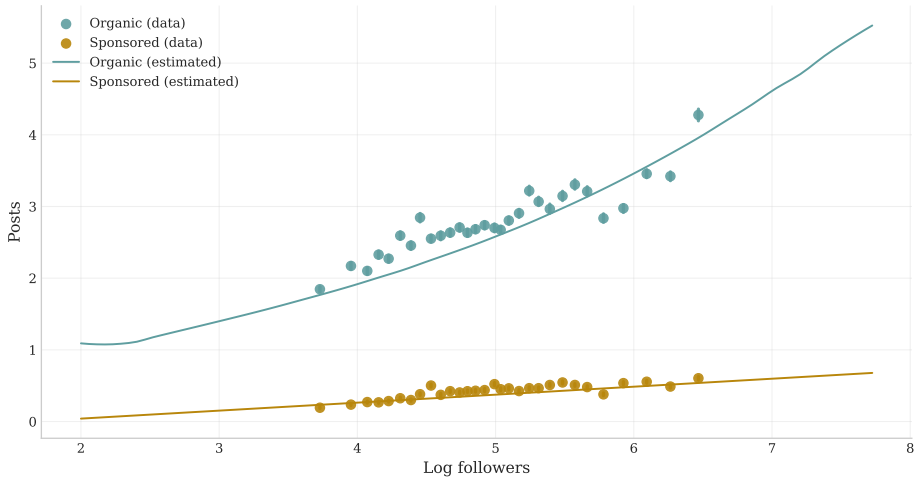
Table 7.1: Estimated parameter values

Parameter	Value	Standard error	Source
π_0	0.206	0.024	Regression
π_1	0.489	0.006	Regression
τ_o	0.00019	0.00001	Regression
τ_s	0.00010	0.00002	Regression
τ_ℓ	0.00028	0.00002	Regression
σ_f	0.0079202	NA	Regression
β	0.99	NA	Assumed
η	6	NA	Assumed
θ_e	0.00344230635	0.00028947	Estimated
θ_m	10.1007941	0.28487783	Estimated

Figure 7.1 plots the optimal policy generated by the estimated parameter values: the number of organic and sponsored posts a lifetime-income-maximizing influencer produces for each possible follower count. The plot also includes a binscatter of the observed numbers of posts in the data. The estimated optimal policy closely matches observed choices. An influencer with 10,000 followers optimally produces about two organic posts per week and about 0.25 sponsored posts per week (one per month). For an influencer with 1,000,000 followers, these numbers increase to 3.5 and 0.5, respectively.

Since pay for sponsored content increases with follower count, it is not surprising that both in the data and according to the optimal policy, influencers with more followers produce more sponsored posts. Organic

Figure 7.1: Optimal policy vs data (estimated parameters)



posts, on the other hand, generate no direct income for the influencer, but larger influencers make more organic posts than smaller influencers. In my model, given the follower count transition function (Equation 3.4), two influencers with different follower counts receive the same *percentage* increase in followers when they produce one additional organic post. This means the larger influencer sees a larger increase in raw follower count, which generates a larger increase in pay per sponsored post. The influencer with a higher follower count therefore sees a higher future payoff from producing an organic post. Since influencers with more followers have larger incentives to produce organic posts, the optimal policy slopes upwards. Intuitively, content from an influencer with more followers can spread through a larger social network, generating a larger increase in the influencer's follower count.

I have established an influencer's optimal posting behavior, which closely matches observed data. Next, I use my quantitative model to examine some popular beliefs about influencer behavior and influencer careers.

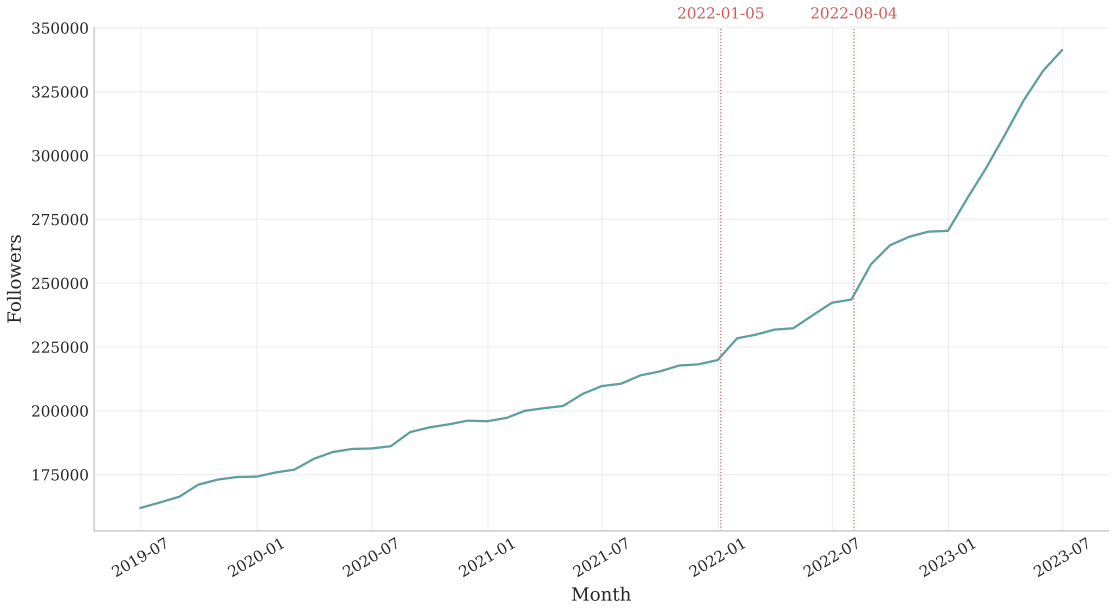
8 The influencer's career

The influencer economy remains poorly understood, and the public holds a wide variety of beliefs about the nebulous job title "influencer". Theoretical literature also suggests mechanisms that have not yet been checked empirically. In this section, I analyze four existing ideas about influencer behavior through the lens of my model. While the list of theories I explore is not exhaustive, my analysis demonstrates the utility of my quantitative model of influencers. Together, the four patterns I explore characterize a typical influencer's career were they to follow the optimal policy I estimate.

Viral posts. "Viral posts" are an important part of content on social media, and they often impact broader culture. For their creators, they can act as a springboard to fame. "Tube Girl", for example, was largely unknown before she created viral posts that ultimately led to a Met Gala invitation (Kessler, 2023). These and similar stories suggest that a single sufficiently viral post can catapult an influencer to fame. Whether this is really true on a broader scale is an empirical question, and I can use my sample of influencers and my

estimated model to investigate it. As an initial empirical example, Figure 8.1 shows follower count history

Figure 8.1: Followers over time for dadaeats



Note: Red lines indicate viral video posts.

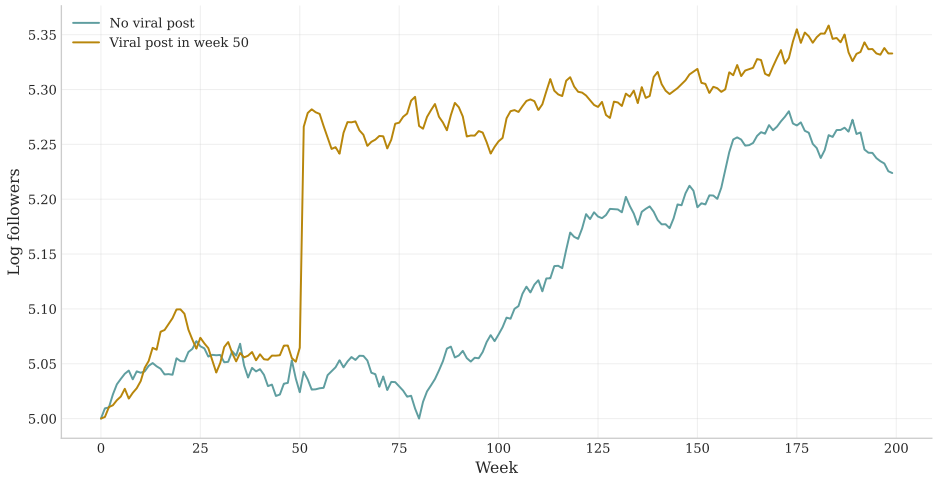
for "@dadaeats", a recipe creator with 367,983 Instagram followers as of October 3rd, 2023. There are periods of faster growth, which sometimes follow viral posts.[10] These appear to alternate with periods of slower growth. This pattern suggests that while an influencer might get a short-lived boost in growth from a viral post, growth will eventually return to its previous pace, and the impact of the viral post does not persist forever. This temporary boost in growth aligns with network-based diffusion. The viral post brings in many new followers quickly. These new followers then spread information about the influencer to their social connections, generating continued growth. Eventually this process slows and growth returns to its pre-viral-post trend.

I can simulate viral posts in my model through the ϵ_t term in Equation 3.4. While I do not directly micro-found viral posts, I can introduce them through a large, positive follower count shock. Then, I can analyze how the influencer's optimal behavior and follower count change over time in response. Figure 8.2 shows the result of such an experiment, in which I simulate the model for 200 weeks for two influencers who both start with 100,000 followers. In week 50 of the simulation, one influencer receives a standard follower count shock from the distribution $\mathcal{N}(0, \sigma_f^2)$. The other influencer receives an artificial, large increase of about 68,000 followers in week 50. While the shock, by construction, increases follower count, it does not change the *slope* of the influencer's follower count curve over time. It only shifts it up, at which point the curve returns to its pre-shock trend. The gap between the two influencers in the plot persists, although it closes somewhat due to different random shocks to follower counts over time.

In my model, an influencer can increase the size of their audience by posting more. Viral posts provide only a transient boost, which means that posting consistently is the primary driver of growth. Anecdotal

[10] I classify viral video posts as those with a view count more than three standard deviations above the mean view count of the influencer's previous posts.

Figure 8.2: Follower growth over time



evidence on social media platforms aligns with this story, where creators complain that a short break in posting can lead to subsequent content receiving less attention.[11] In general, my model does not support the idea that a single viral post is sufficient for growth on a social media platform. This conclusion informs popular conceptions of social media and suggests that luck may play only a small role in an influencer's career.

Earnings. Nearly 25% of Gen Z would like to pursue content creation as a career (Novacic, 2019). My estimated model allows me to evaluate the viability of this career, since it delivers the optimal number of posts an influencer should produce along with the pay the influencer earns. To compare content creation to other careers, I also need a measure of the amount of time an influencer spends making one post. This affects the hourly wage an influencer can earn, which offers a useful point of comparison to other careers. I fill this

Table 8.1: Earnings by follower count

	Posts per week	Hours per week	Earnings per week ($)		Earnings per hour ($)		Hours per post
Followers	Model	HypeAuditor	Model	HypeAuditor	Model	HypeAuditor	Both
10,000	2.18	27	38.27	327.69	1.42	12.14	12.38
50,000	2.71	28	108.90	475.62	3.89	16.99	10.35
500,000	3.62	30.6	445.16	811.62	14.55	26.52	8.45
1,000,000	3.95	32	671.02	1,349.31	20.97	42.17	8.11

Note: HypeAuditor income and hours use the upper endpoints of the ranges reported in the survey: 1,000–10,000, 10,000–50,000, 50,000–500,000, and 500,000–1,000,000.

gap using a 2021 survey from HypeAuditor, which asked influencers about their work hours and income (Baklanov, 2021). The survey reports the number of hours per week influencers spend creating content, for various follower counts. I combine these numbers with two separate estimates of earnings per week to calculate an estimate of hourly wages at different follower counts. The first estimate comes from the optimal policy implied by my estimated model: earnings per week equals sponsored posts per week times pay per sponsored post. The second estimate is survey-reported earnings per month times 12/52.

[11] See, for example, https://www.reddit.com/r/InstagramMarketing/comments/19dzxps.

Table 8.1 reports the estimates. The survey data imply higher hourly earnings than does the optimal policy from my model. The discrepancy could have several sources. My model only accounts for revenue from one platform, but influencers sometimes have other sources of income, like Patreon, and the survey does not specifically ask about how each participant arrived at their earnings estimate. A second possibility is that influencers in the survey do not fully account for the dynamic effects of their posting choices. Sponsored posts generate slightly slower follower growth according to my estimates; if influencers do not fully account for this penalty, they would produce more sponsored posts than the optimal number my model implies, and their earnings would be higher. Finally, there could be selection into survey participation; the HypeAuditor numbers might reflect particularly successful content creators, or the surveyed influencers might use particularly high-earning months to come up with their answers. My goal is to broadly characterize earnings, so I do not take a stance on which numbers are more accurate.

According to both my model and the survey, influencer wages are not astronomically high. An influencer with 50,000 followers earns below minimum wage in some states. An order of magnitude increase in follower count yields a wage comparable to emergency medical technicians ($19.88 per hour; U.S. Bureau of Labor Statistics, 2025). At 1,000,000 followers, influencers earn similar wages to accountants ($39.27 per hour). For reference, "Tube Girl" has about 920,000 followers across both Instagram and TikTok. These earnings numbers exclude the costs of content production, like camera equipment or subscriptions to video production software, that do not apply directly to hourly-wage workers in other occupations. Therefore, true influencer wages might be even lower than the reported estimates. While my analysis above is not an in-depth investigation of the labor market for content creators, it puts the job in context relative to other occupations, and largely suggests that content creation is not a "get-rich-quick" scheme.

Production costs. In addition to measuring influencer revenue, the HypeAuditor survey and my model together offer a proxy for the cost of producing a post. Since I observe posts per week (from my model) and hours per week (from the survey), I can calculate the number of hours influencers of different sizes spend producing one post. Table 8.1 reports some example values. Hours per post decline as follower count increases. This might happen because influencers gain experience and learn to make content more quickly. On the other hand, larger influencers sometimes hire dedicated content production teams, in which case total human work hours spent on the post might stay the same or increase, while the individual influencer spends less time since they offload some of the work to employees (Scanlon, 2023).

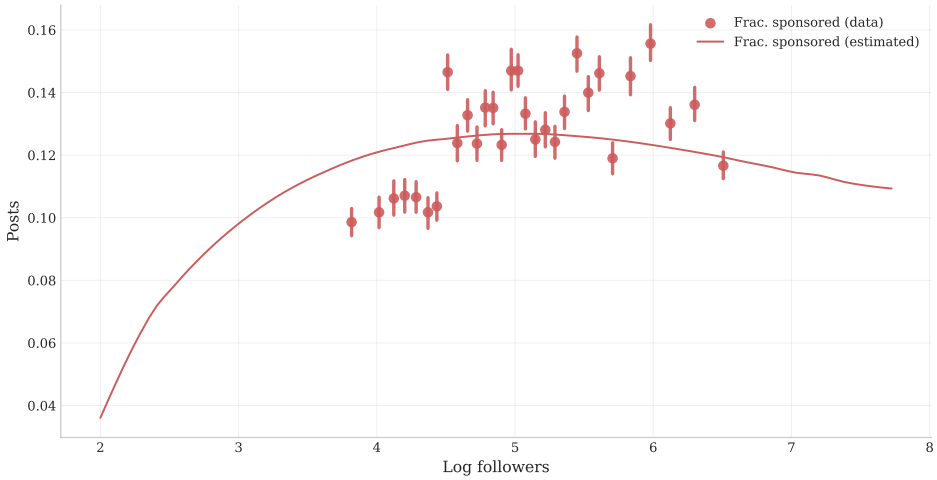
In my model of influencer behavior, the terms c_e and c_m (Section 3) capture the costs of producing content. I assume that these cost curves do not depend directly on follower count. Testing this assumption is an interesting direction for future work. The HypeAuditor data do not rule out my assumption, since I do not observe the size of influencers' content production teams.

Investing and harvesting. Existing theoretical models of influencers predict distinct periods of "investing" and "harvesting" (Mitchell, 2021; Nistor et al., 2024). In the investing period, an influencer avoids making sponsored content in order to build a positive reputation. After building a sufficiently strong reputation, the influencer leverages it to earn revenue by making sponsored content.

I analyze investing and harvesting in my context using the *fraction* of sponsored content an influencer produces. This choice facilitates comparisons to existing theoretical results that focus on measures of "authenticity". Mitchell (2021) considers the intensity of an ad technology as a number in $[0, 1]$, while Nistor et al. (2024) allows the influencers to either accept 100% of advertising offers, or only some fraction of them

that are sufficiently high quality. The raw number of sponsored posts does not map directly to authenticity, since the effect of an additional sponsored post on authenticity is ambiguous if the influencer simultaneously makes more organic posts. Both existing models predict a distinct point at which the influencer switches from investing to harvesting.

Figure 8.3: Optimal policy (frac. sponsored) vs. data



Note: Frac. sponsored (data) is a binscatter of the fraction of sponsored posts at each observed follower count. Bars represent bootstrapped 95% confidence intervals around the mean.

Figure 8.3 plots the fraction of sponsored posts as a function of follower count, both in the data and from an influencer following the estimated optimal policy. In the data, the mean fraction increases above about 30,000 followers. However, the increase is small, from about 10% to 12%-16%, and the increase in the mean from one bin to the next is partly driven by which influencers appear in that bin. The literature described above, on the other hand, derives a dramatic switch from a low-advertising state to high advertising.

According to the optimal policy, an influencer increases their fraction of sponsored content up to about 100,000 followers, and then decreases it slowly. This pattern occurs because sponsored content incurs only a small follower growth penalty. Sponsored content becomes more attractive as the influencer grows because more followers generate larger payments, which outweigh the additional cost of producing more sponsored posts. As described in Section 7.3, posting generates a percentage increase in followers, which means that organic posts increase follower count more if the influencer already has more followers. Organic posts therefore become more attractive for sufficiently large influencers, so the fraction of sponsored posts begins to decline. The non-monotonicity in the optimal policy might suggest alternating periods of investing and harvesting, which also occur in the model of Mitchell (2021).

These results suggest that influencers do not suddenly switch from investing to harvesting at some threshold follower count. Instead, the relatively smooth adjustment of the fraction of sponsored content with follower count is more consistent with the extension in Nistor et al. (2024) in which influencers can continuously choose a fraction of sponsorship offers to accept.

In this section, I used my estimated model to describe several stylized facts about how an influencer's career evolves over time. My analyses hold key model parameters fixed, but policy changes and technological innovations might change an influencer's incentives and alter the trajectory of their career. To examine such

changes in more detail, I now turn to counterfactual analyses.

9 Counterfactuals

With my estimated model in hand, I can vary certain parameters to understand how influencer behavior would change if the content production environment changed. First, I consider a counterfactual world in which followers react more negatively to sponsored content; this might represent an influencer operating on a different platform whose users have different preferences for advertising. Second, I simulate an increase in the cost of producing sponsored content, which could result from policy changes that increase regulation of influencer marketing. For each scenario, I show how the influencer's optimal policy changes, and I use the marginal benefit and marginal cost of posting to explain where the change comes from.

9.1 Marginal cost and marginal benefit of posting

Before analyzing specific counterfactual scenarios, I briefly detail how I calculate the marginal costs and benefits of organic and sponsored posts. These calculations are valuable for understanding why the optimal policy shifts, or does not shift, under specific parameter changes.

Organic posts. The marginal cost of an organic post is directly related to the effort cost function c_e, which is the only component of current-period utility that depends on the number of organic posts the influencer produces. I can calculate it exactly as

$$\frac{\partial}{\partial o} c_e(o, s) = \theta_e \eta (o + s)^{\eta - 1} \tag{9.1}$$

The marginal benefit of an organic post comes from the increase in future follower count it generates, which yields future revenue. It is given by the derivative of the expected next-period value function, which I approximate numerically:

$$\frac{\partial}{\partial o} \left[\beta \int_2^9 V(x) \phi(x; f, o, s, \sigma_f) \, dx \right] \tag{9.2}$$

Sponsored posts. The marginal cost of a sponsored post also derives from the current-period production cost functions:

$$\frac{\partial}{\partial s} \left[c_e(o, s) + c_m(s) \right] = \theta_e \eta (o + s)^{\eta - 1} + \theta_m e^{\theta_m s} \tag{9.3}$$

Sponsored posts, like organic posts, increase the influencer's future follower count. They also generate current-period revenue through payments from the sponsoring brands. The marginal benefit is therefore given by the following expression:

$$\frac{\partial}{\partial s} \left[\alpha(f) s + \beta \int_2^9 V(x) \phi(x; f, o, s, \sigma_f) \, dx \right] \tag{9.4}$$

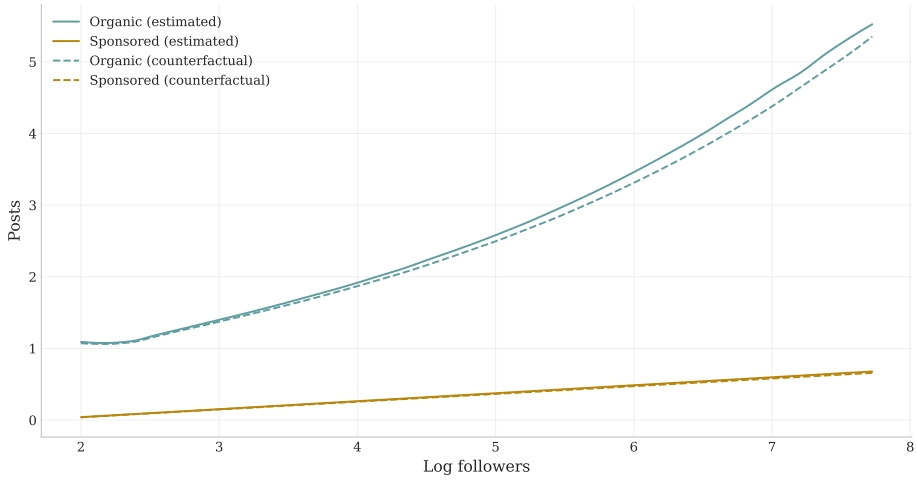
Again, I numerically approximate the derivative of the value function.

9.2 Negative effect of sponsored posts on follower growth

First, I analyze a change in how the influencer's audience responds to advertising by changing the parameter τ_s that governs the effect of a sponsored post on follower growth. In the baseline model, τ_s is positive, so on

net, sponsored posts increase the size of the influencer's audience. In this counterfactual I set $\tau_s = -0.00194$ so that sponsored posts generate a net loss of followers. The specific value comes from the estimate in Cheng and Zhang (2022), who estimate a negative effect of sponsored content on YouTube. This parameter change could reflect a shift in the preferences of the influencer's audience to be less amenable to advertising. Alternatively, it might represent an influencer operating on a different platform where sponsored content has different effects. If Instagram, for example, is viewed as a platform for product discovery, while YouTube is largely for entertainment, then audiences on YouTube might react more negatively to advertising.[12] Figure

Figure 9.1: Optimal policy vs data: dynamic cost for sponsored content



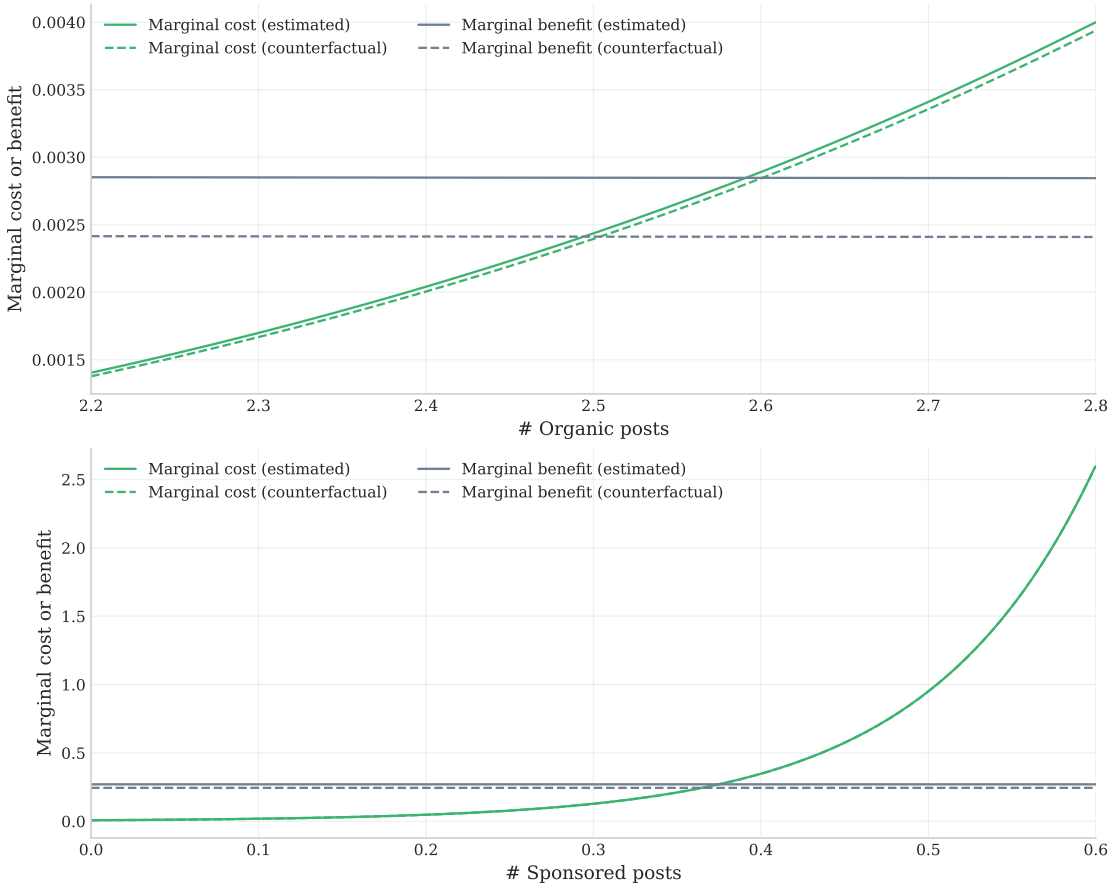
9.1 plots the optimal policy under the altered value of τ_s. Compared with the baseline optimal policy, the negative impact of sponsored posts on follower growth has only a small impact on overall content production. For an influencer with 100,000 followers, the optimal number of organic posts drops from 2.581 to 2.495, and sponsored posts decline from 0.375 to 0.365.

To understand the change in the optimal policy, I calculate the marginal benefits and marginal costs of organic and sponsored posts under the original and new values of τ_s. Figure 9.2 plots each curve as a function of the number of posts of each type. For each curve, I fix follower count at 100,000. In the organic post plot, I fix sponsored posts at the optimal number for an influencer with 100,000 followers under $\tau_s = 0.0001$ and $\tau_s = -0.00194$ for the estimated and counterfactual curves, respectively. The optimal number of sponsored posts at 100,000 followers is lower under the counterfactual τ_s, so the marginal cost of an organic post decreases slightly (Equation 9.1). The marginal benefit of an organic post also declines. Under a negative value of τ_s, sponsored posts have a larger cost in terms of future follower count. Making one more organic post yields followers in the future, but monetizing those followers through sponsored posts is now more costly, so the organic post has a lower return for the influencer.

The marginal benefit of sponsored posts also declines, since they decrease the influencer's next-period follower count. Pay for a sponsored post outweighs the loss of future followers, so the marginal benefit remains positive. The marginal cost curve shifts down very slightly because the optimal number of organic

[12] This is one possible reason I estimate a net positive effect of sponsored posts on Instagram, while Cheng and Zhang (2022) estimate a negative effect on YouTube.

Figure 9.2: Marginal cost and marginal benefit: $\tau_s = -0.00194$



posts under the counterfactual τ_s is lower (Equation 9.3). The influencer's optimal choice of sponsored posts does not change substantially because the marginal cost curve is steep.

The decrease in the marginal benefit of organic posts highlights the dynamic tradeoff to sponsored content. Making more sponsored posts leads to a lower future follower count, relative to the case where those posts were organic. Since organic posts increase follower count more for influencers with more followers, sponsored posts ultimately reduce the return to organic posts.

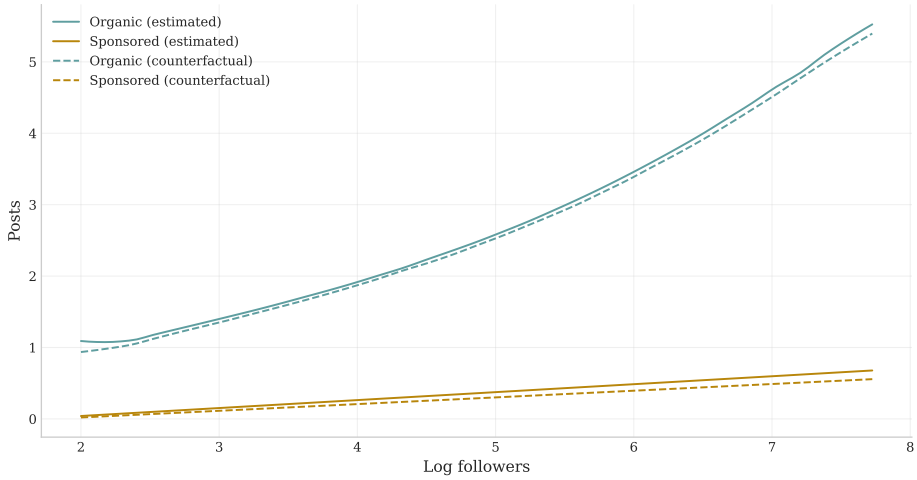
9.3 Increased cost of sponsored content

In my second counterfactual exercise, I simulate increasing the stringency of influencer advertising regulation. The Federal Trade Commission currently requires influencers to clearly disclose all sponsored posts, but improper disclosure appears to be common (Ershov et al., 2023). More aggressive monitoring and regulation could benefit consumers by clarifying whether influencers' product recommendations are genuine. On the other hand, producing sponsored posts might become more costly for influencers. For example, the FTC might require influencers to log their brand partnerships in a database, so sponsored content would take more time to make. Since influencers primarily earn revenue through sponsored posts, making them less profitable could reduce the incentives to make organic posts (Mitchell, 2021). If consumers enjoy seeing organic posts, more costly sponsored content could reduce their welfare. The net effect of sponsored content

regulation is therefore ambiguous.

While I cannot quantify consumer welfare directly, I use my model to investigate the importance of the second channel, where regulation reduces the incentives to produce organic content. I assume regulation would increase the cost c_m of producing a sponsored post, so I increase the value of θ_m from 10.10 to 12, which roughly triples the marginal cost of a sponsored post for an influencer with 100,000 followers. Figure 9.3 plots the optimal policy under the baseline and counterfactual values of θ_m. A creator with

Figure 9.3: Optimal policy vs data: increased cost of sponsored content

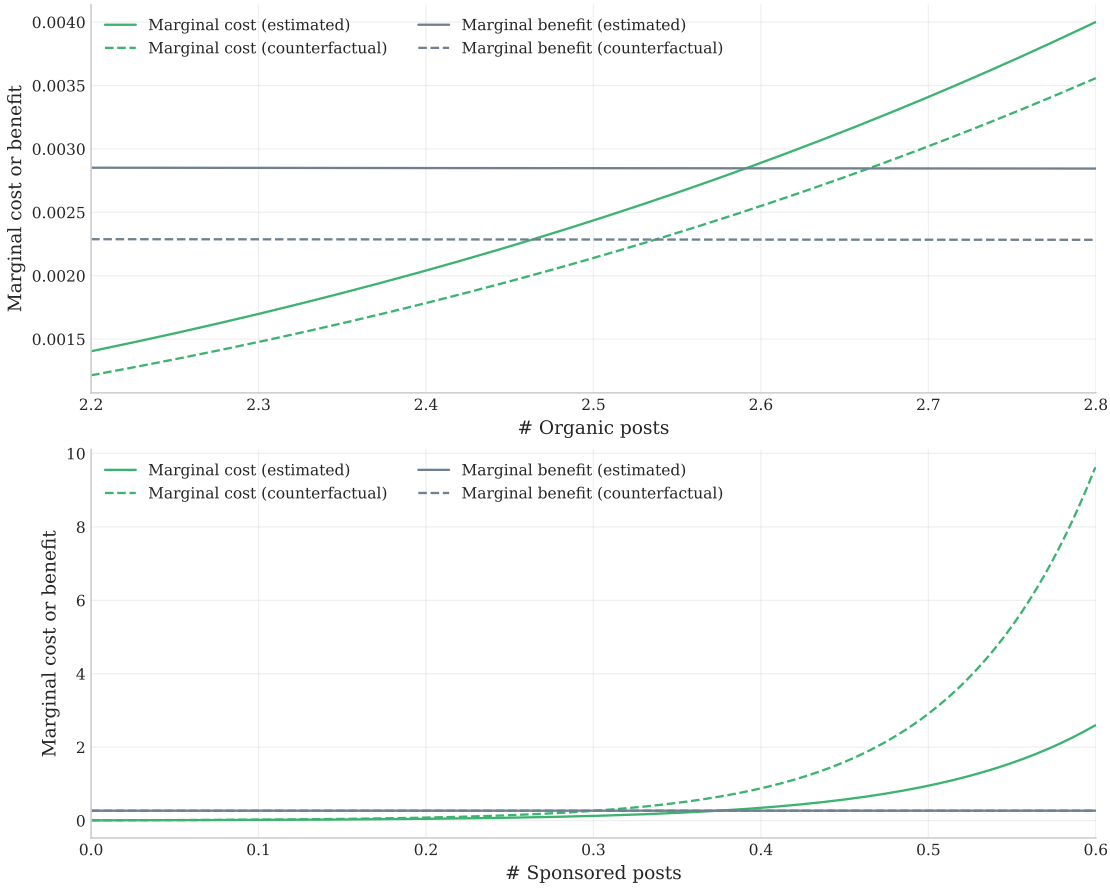


100,000 followers produces about 0.4 sponsored posts per week at baseline and about 0.3 per week under the counterfactual scenario, a 25% reduction. Organic content, however, falls only slightly, from about 2.6 posts per week to about 2.55.

Figure 9.4 plots the marginal benefit and marginal cost curves for an influencer with 100,000 followers under the baseline and counterfactual parameter values, analogously to Figure 9.2. The marginal cost curve for sponsored content shifts upward because of the increase in θ_m. The marginal benefit of sponsored posts decreases, but only very slightly, due to the lower future profitability of sponsored posts. Similar to the previous counterfactual, organic posts benefit the influencer only through future sponsored posts, which are now more costly, so the marginal benefit of an organic post decreases. On the other hand, since sponsored content production decreases due to its higher marginal cost, the influencer's optimal choice moves leftward along the effort cost curve c_e, so the marginal cost of an organic post decreases. These two effects nearly cancel out, and the net effect is a slight decrease in the number of organic posts produced.

The simulated policy change decreases influencer earnings since they make fewer sponsored posts and profit less from each one. The impact on consumer welfare depends on preferences for organic versus sponsored content, since consumers see fewer sponsored posts but also fewer organic posts. For influencers, the increased cost to make sponsored posts does reduce the incentives to make organic posts. However, influencers also optimally shift some of their effort away from sponsored content and towards organic content. The second effect works in opposition to the first, yielding a small net effect on organic content production. This mechanism arises because I explicitly model content production in a way that allows influencers to substitute effort between post types.

Figure 9.4: Marginal cost/marginal benefit: $\theta_m = 12$



10 Conclusion

The creator economy exercises broad influence over consumers, but the careers of "influencers" remain poorly understood. Influencers face a tradeoff: they can use advertising to earn money immediately, or they can grow their audience and earn higher payments in the future. This tradeoff remains largely empirically untested. I develop a dynamic model of influencer content choice in which organic and sponsored posts affect the growth of the influencer's audience and thereby alter future income.

I estimate the model's parameters using data on follower counts, posting activity, and payments for 1,369 influencers. While my data span a range of follower counts, I observe neither the largest influencers nor the smallest. Future work could investigate the careers of influencers in these tails of the distribution.

Using the data, I quantify the effect of sponsorship on engagement by comparing identical posts across two platforms, and I show that about half of the negative effect of sponsorship comes from reduced effort on sponsored posts, and about half comes from audience aversion to advertising. While my empirical strategy builds on existing literature, more work is needed to fully identify how audiences respond to sponsored posts.

Next, I estimate how organic and sponsored posts affect the size of an influencer's audience, and I show that sponsored posts generate positive but slightly slower growth. I then estimate the parameters of the cost function for producing content by comparing simulated influencer careers to the data. Since sponsored posts have only a small dynamic cost, it appears that these short-term production costs drive the observed

difference between organic and sponsored post production.

Finally, I simulate the effect of making sponsored posts more costly to produce, for example, by increasing disclosure requirements. I show that such a change, sufficiently large to reduce sponsored content production by 25%, decreases organic post production by just 2.7%, in part because influencers substitute some of the effort they previously spent on sponsored posts to making organic posts. The concern that regulating sponsored content would disincentivize organic content production appears unlikely.

References

Bairathi, M. and Lambrecht, A. (2023). Influencer marketing: Sponsorship disclosure and authenticity. *Available at SSRN 4679776*.

Baklanov, N. (2021). Our survey data of 1865 instagram influencers.

Berger, J. and Milkman, K. L. (2012). What makes online content viral? *Journal of marketing research*, 49(2):192–205.

Cascio Rizzo, G. L., Villarroel Ordenes, F., Pozharliev, R., De Angelis, M., and Costabile, M. (2024). How high-arousal language shapes micro-versus macro-influencers' impact. *Journal of Marketing*, page 00222429231207636.

Cheng, M. M. and Zhang, S. (2022). Reputation burning: Analyzing the impact of brand sponsorship on social influencers. *Available at SSRN 4071188*.

Ershov, D., He, Y., and Seiler, S. (2023). How much influencer marketing is undisclosed? evidence from twitter.

Ershov, D. and Mitchell, M. (2020). The effects of influencer advertising disclosure regulations: Evidence from instagram. In *Proceedings of the 21st ACM Conference on Economics and Computation*, pages 73–74.

FYPM (2024). Frequently asked questions.

Geyser, W. (2026). Instagram influencer rates: A complete guide for brands in 2026.

Giuffredi-Kähr, A., Petrova, A., and Malär, L. (2022). Sponsorship disclosure of influencers–a curse or a blessing? *Journal of Interactive Marketing*, 57(1):18–34.

Gong, Z. (2021). Growing influence. Technical report, working paper, University of Toronto.

Hughes, C., Swaminathan, V., and Brooks, G. (2019). Driving brand engagement through online social influencers: An empirical investigation of sponsored blogging campaigns. *Journal of marketing*, 83(5):78–96.

Judd, K. L. (1998). *Numerical methods in economics*. MIT press.

Kessler, A. (2023). How TikTok star "Tube Girl" became a fashion it-girl overnight.

Leung, F. F., Gu, F. F., Li, Y., Zhang, J. Z., and Palmatier, R. W. (2022). Influencer marketing effectiveness. *Journal of marketing*, 86(6):93–115.

Li, I. (2023). Sponsorships in livestreaming: Monetization and disclosure behavior of influencers on twitch.

Li, W., Liu, H., Yang, F., and Yao, R. (2016). Housing over time and over the life cycle: A structural estimation. *International Economic Review*, 57(4):1237–1260.

Mitchell, M. (2021). Free ad (vice): internet influencers and disclosure regulation. *The RAND Journal of Economics*, 52(1):3–21.

Nelson, P. (1974). Advertising as information. *Journal of political economy*, 82(4):729–754.

Nistor, C., Selove, M., and Villas-Boas, J. M. (2024). Influencers: To grow or to monetize. *Available at SSRN 4708107*.

Novacic, I. (2019). "it's kinda crazy": Kid influencers make big money on social media, and few rules apply.

Pei, A. and Mayzlin, D. (2022). Influencing social media influencers through affiliation. *Marketing Science*, 41(3):593–615.

Řehůřek, R. and Sojka, P. (2010). Software Framework for Topic Modelling with Large Corpora. In *Proceedings of the LREC 2010 Workshop on New Challenges for NLP Frameworks*, pages 45–50, Valletta, Malta. ELRA. http://is.muni.cz/publication/884893/en.

Sahni, N. S. and Nair, H. S. (2020). Does advertising serve as a signal? evidence from a field experiment in mobile search. *The Review of Economic Studies*, 87(3):1529–1564.

Scanlon, K. (2023). It takes a village: A look at the hidden teams behind successful creators.

Statista (2024a). Daily time spent on social networking by internet users worldwide from 2012 to 2024.

Statista (2024b). Social media advertising.

Tang, Q., Gu, B., and Whinston, A. B. (2012). Content contribution for revenue sharing and reputation in social media: A dynamic structural model. *Journal of Management Information Systems*, 29(2):41–76.

Tian, Z., Dew, R., and Iyengar, R. (2024). Mega or micro? influencer selection using follower elasticity. *Journal of Marketing Research*, 61(3):472–495.

U.S. Bureau of Labor Statistics (2025). Occupational employment and wages—may 2024.

Valsesia, F., Proserpio, D., and Nunes, J. C. (2020). The positive effect of not following others on social media. *Journal of Marketing Research*, 57(6):1152–1168.

Wies, S., Bleier, A., and Edeling, A. (2023). Finding goldilocks influencers: How follower count drives social media engagement. *Journal of Marketing*, 87(3):383–405.

Appendix A Influencer pay

Figure A.1: Pay vs follower count

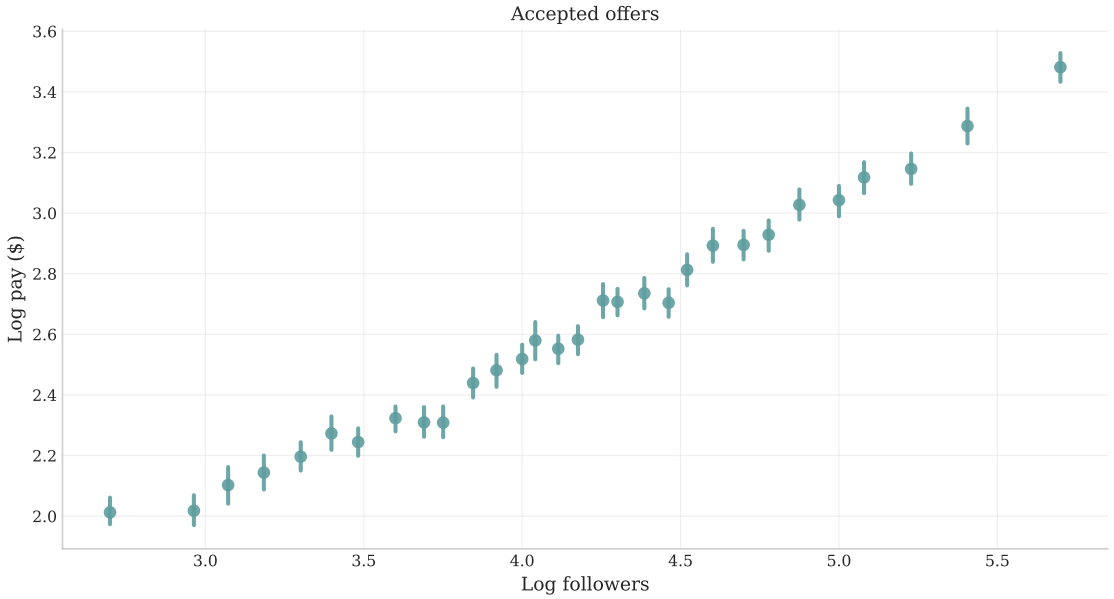


Figure A.2: Pay per follower vs follower count

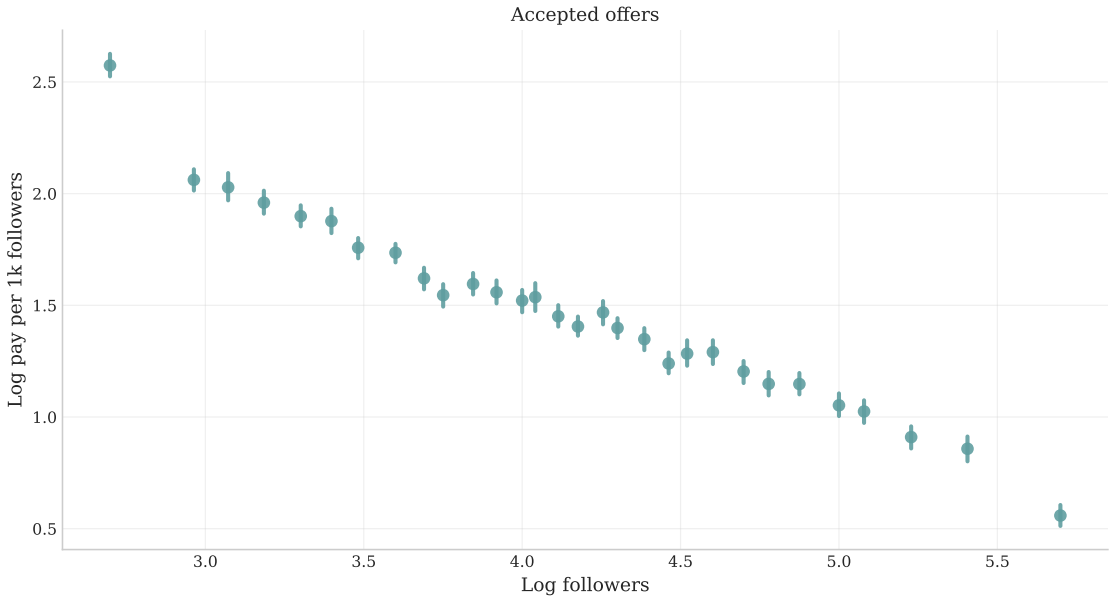


Figure A.3: Review of accepted collaboration

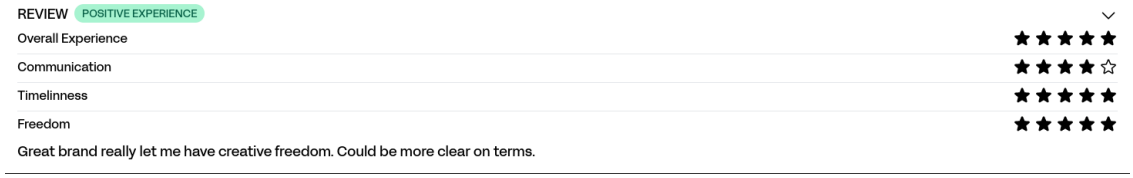


Figure A.4: Pay vs deliverables, accepted offers only

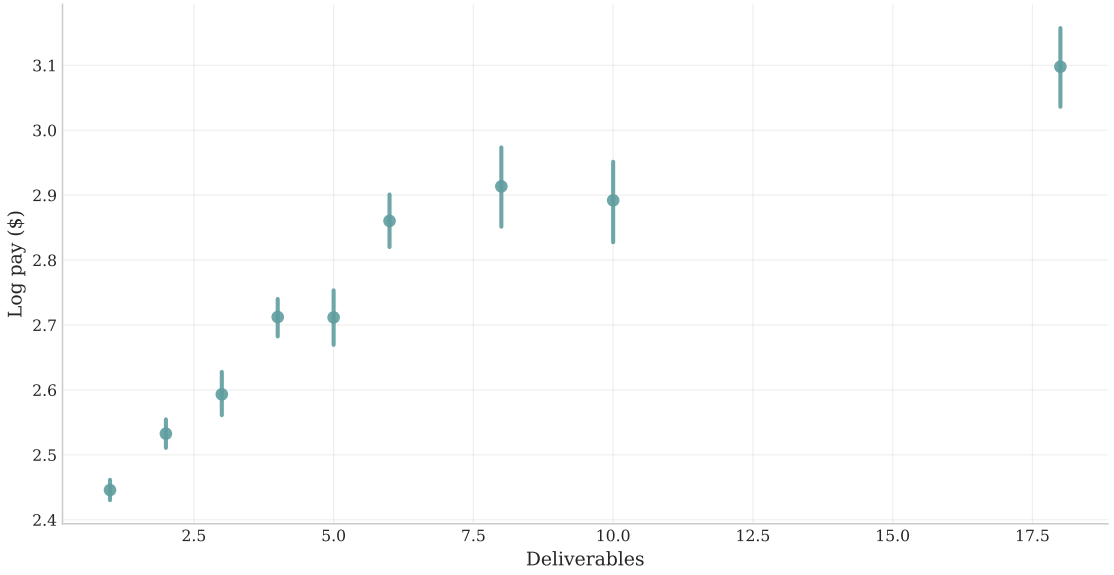


Table A.1: Dependence of pay on Instagram follower count

	Log pay per deliverable	Log pay per deliverable
Log Instagram followers	0.509***	0.502***
	(0.007)	(0.007)
Intercept	0.114***	0.607
	(0.031)	(0.414)
Date FE	No	Yes
N	11,044	11,044
R2	0.298	0.311

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

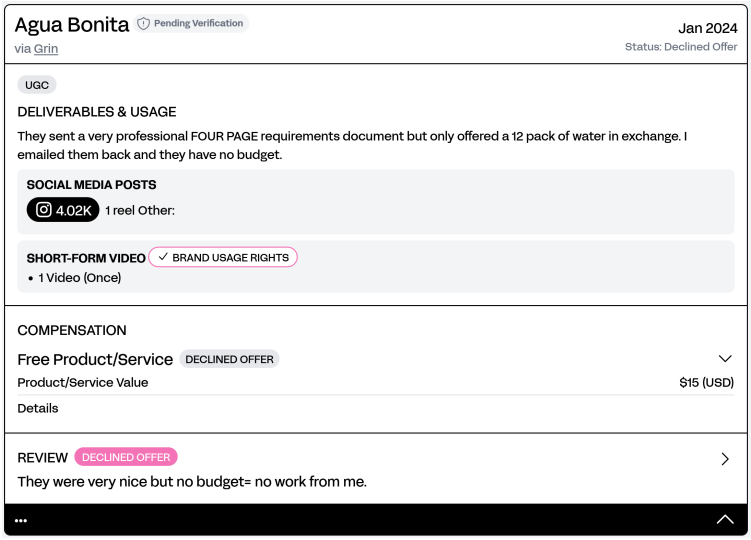
Table A.2: Currency conversion rates used to convert pay to USD

Currency	USD equivalent
USD	1.00
CAD	0.73
GBP	1.28
EUR	1.08
AUD	0.67

A.1 Declined reviews

In my sample, 809 reviews are marked "Declined Offer", which means a brand proposed a sponsorship to an influencer and the influencer said no. Figure A.5 shows an example in which the influencer declined because the compensation (free water) was insufficient. Reading through the reviews, I also found several

Figure A.5: Review scores



in which the influencer declined because the brand was a poor fit for their audience. Table A.3 predicts the probability of declining given several review characteristics. Influencers with more followers are more likely to decline offers, probably because they receive more offers and can be pickier. Pay and the influencer's "overall experience" with the brand have a significant negative impact on the probability of declining the offer, as expected. Freedom, the level of creative control the influencer has over the sponsored post, increases the probability of accepting an offer. In Pei and Mayzlin (2022), a firm that affiliates too closely with an influencer can reduce the persuasiveness of that influencer's product review. The result in Table A.3 suggests that influencers understand this cost.

Appendix B Sponsored post classification

B.1 Alternative classifications of sponsored posts

There are many ways to separate sponsored from organic posts. Rather than claim a perfect classification system, I implement some other methods to provide upper and lower bounds on the number of sponsored posts in my data. I start with the least restrictive classification: assuming any post that mentions another Instagram account is sponsored. This creates false positives because influencers sometimes mention their friends or other influencers, and these mentions do not indicate sponsorship. On the other hand, Figure 5.2 demonstrates that a mention is sometimes the only evidence of sponsorship in the post text. My primary classification method only looks for mentions of brands in the FYPM.vip data; treating all mentions as evidence of sponsorship ensures no brand is excluded. It therefore provides an upper bound on the number of sponsored posts in the data. Table B.3 compares organic and sponsored Instagram posts, where any

Table A.3: Predicting declined offers

	Declined
Log followers	0.536***
Log pay	-1.040***
Free product	0.234+
Log deliverables	0.111
Usage rights	0.192+
Verified	-0.139
Overall experience (1-5)	-1.636***
Communication (1-5)	0.386***
Timeliness (1-5)	0.318***
Freedom (1-5)	-0.135**
# Words	-0.004***
Intercept	0.260
N	15,559
N (declined)	512

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

post with a mention other than the influencer's own username is classified as sponsored. Unsurprisingly, the number of sponsored posts increases significantly.

Some, but not all, posts mention brands from the FYPM.vip data. To gauge how well FYPM covers the universe of possible sponsors, I try classifying any post mentioning a brand from FYPM as sponsored. Table B.4 describes sponsored posts classified this way. Many of the mentions in Table B.3 are either false positives or mention brands not in the FYPM data.

Checking for explicit sponsorship disclosure should almost completely avoid false positives. It seems unlikely that "#sponsored" would appear in an organic post, so 141,747 is probably a lower bound on the number of sponsored posts in my data. Table B.5 classifies any post using the paid partnership label or hashtags like "#ad" as sponsored. Table B.6 only looks for the paid partnership label. There is a remote possibility that some influencers use disclosure like "#ad" on posts that are not sponsored so they appear to receive more sponsorship offers than they actually do. The paid partnership label cannot be abused this way because the partnering brand also participates in the disclosure. Note that my primary classification method in Table 5.1 is the union of those in Tables B.4 and B.5.

Finally, I apply sponsorship classification methods from Ershov and Mitchell (2020). Ideally I would use their machine learning algorithm to detect sponsored content, but it relies on a reliable ground truth. The authors use Instagram posts after a regulation change in Germany which required disclosure on sponsored posts. Fines and enforcement followed the change, so Ershov and Mitchell (2020) assume all sponsored posts are disclosed after the new regulations. Their German Instagram data therefore serves as training data for machine learning models that differentiate organic and sponsored posts. Since I used United States data, I have no equivalent training dataset. In fact, Ershov et al. (2023) suggests that most sponsored content is undisclosed.

Table B.1: Ershov and Mitchell (2020) manual "disclosed" classification

Word
#ad
gift
gifted
patron
ambassad
ambassador
collaboration
partnership
collab
sponsored
sponsor
promo
partner
publicit
promotion
advertisement
publi

Instead, I use the manual classification from Ershov and Mitchell (2020). It consists of two lists of words. The first (Table B.1) contains explicit disclosure indicators and the second (Table B.2) consists of words suggesting sponsorship. The second list in particular is broad; it contains words like "until" that could be part of a limited time offer ("coupon valid until October 1st") but could also be present in organic posts. Table B.7 shows that searching text for these words classifies about two-thirds of the Instagram posts in my data as sponsored. Ershov and Mitchell (2020) agree that "our manual definition likely overstates the amount of sponsored content". Searching post text for Ershov and Mitchell (2020)'s explicit sponsorship disclosures yields about 200,000 sponsored posts (Table B.8). This is the closest number to my original classification, which had about 244,000 sponsored posts. Finally, Table B.9 classifies posts as sponsored if they contain a word either from Table B.1 or from Table B.2. Combining the two shows that most posts classified as sponsored because they contain explicit disclosure words also contain other indicators of sponsorship.

Table B.2: Ershov and Mitchell (2020) manual "sponsored" classification

Word
.com
@
available
link
must
have
shop
buy
now
code
%
$
contest
even
launch
tonight
ship
hotel
campa
follow
until
official
video
new
thank
thanks
diet
shake
detox
smoothi
supplement
protein
tea
drink
health

Table B.3: Organic vs sponsored (any mention) Instagram posts

Variable	Organic	Sponsored	p-value
Followers	802,178	904,296	0.000
Likes	10,912	11,034	0.054
# Comments	143	185	0.000
# Words in post	46	68	0.000
Viral	0.062	0.073	0.000
N	1,629,882	1,150,129	NA

Table B.4: Organic vs sponsored (mentions FYPM brand) Instagram posts

Variable	Organic	Sponsored	p-value
Followers	838,346	1,018,910	0.000
Likes	10,819	13,842	0.000
# Comments	155	261	0.000
# Words in post	54	80	0.000
Viral	0.065	0.099	0.000
N	2,646,889	133,122	NA

Table B.5: Organic vs sponsored (disclosed) Instagram posts

Variable	Organic	Sponsored	p-value
Followers	870,738	618,184	0.000
Likes	11,086	8,607	0.000
# Comments	160	167	0.151
# Words in post	53	93	0.000
Viral	0.064	0.11	0.000
N	2,638,264	141,747	NA

Table B.6: Organic vs sponsored (disclosed with platform tool) Instagram posts

Variable	Organic	Sponsored	p-value
Followers	859,804	532,542	0.000
Likes	10,984	9,322	0.000
# Comments	160	199	0.010
# Words in post	54	110	0.000
Viral	0.065	0.15	0.000
N	2,741,827	38,184	NA

Table B.7: Organic vs undisclosed sponsored (Ershov and Mitchell, 2020) Instagram posts

Variable	Organic	Sponsored	p-value
Followers	1,088,040	781,708	0.000
Likes	12,607	10,148	0.000
# Comments	137	172	0.000
# Words in post	19	73	0.000
Viral	0.045	0.077	0.000
N	916,810	1,863,201	NA

Table B.8: Organic vs disclosed sponsored (Ershov and Mitchell, 2020) Instagram posts

Variable	Organic	Sponsored	p-value
Followers	886,096	555,519	0.000
Likes	11,102	9,139	0.000
# Comments	152	267	0.000
# Words in post	52	101	0.000
Viral	0.066	0.079	0.000
N	2,579,343	200,668	NA

Table B.9: Organic vs sponsored (Ershov and Mitchell, 2020) Instagram posts

Variable	Organic	Sponsored	p-value
Followers	1,099,960	780,148	0.000
Likes	12,684	10,129	0.000
# Comments	137	172	0.000
# Words in post	19	73	0.000
Viral	0.045	0.077	0.000
N	902,620	1,877,391	NA

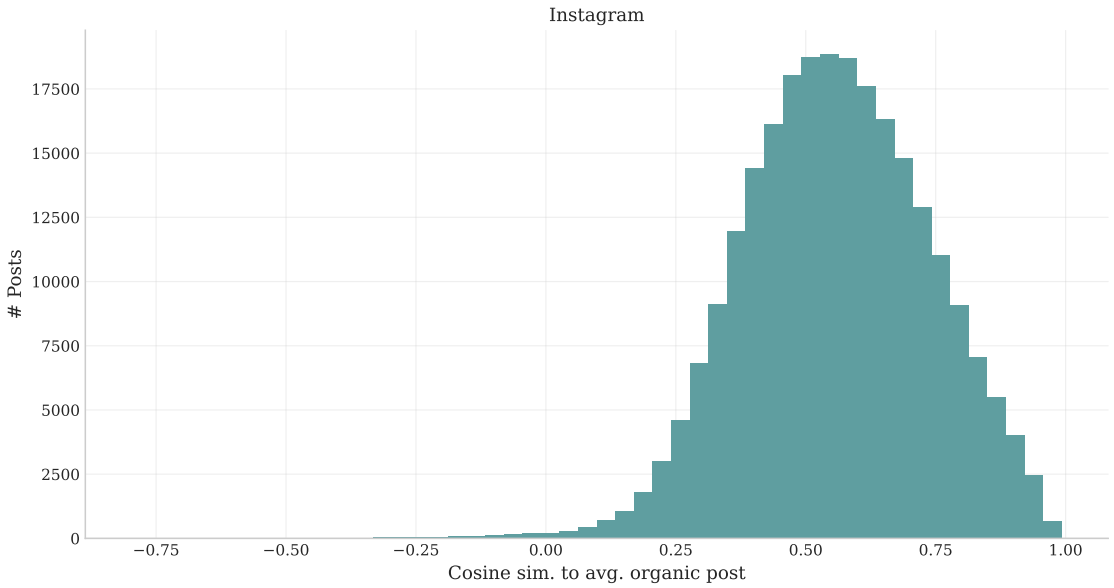
Appendix C Match value

The alignment between a sponsored post and the author's typical content affects the audience's reaction to it (Leung et al., 2022). I calculate a measure of follower-brand fit based on text similarity as follows:

1. Generate vector representations of each post's text using Doc2Vec with 100 features (Řehůřek and Sojka, 2010).
2. Calculate each influencer's "average" organic post on each platform by taking the componentwise mean of all their organic posts.
3. Calculate the match value of a sponsored post as the cosine similarity between the post's vector representation and the influencer's average organic post.

Figure C.1 shows the distribution of match values for sponsored Instagram posts. The median Instagram post has a similarity score of 0.56. Leung et al. (2022) empirically show an inverted U-shaped relationship between

Figure C.1: Similarity to average organic post

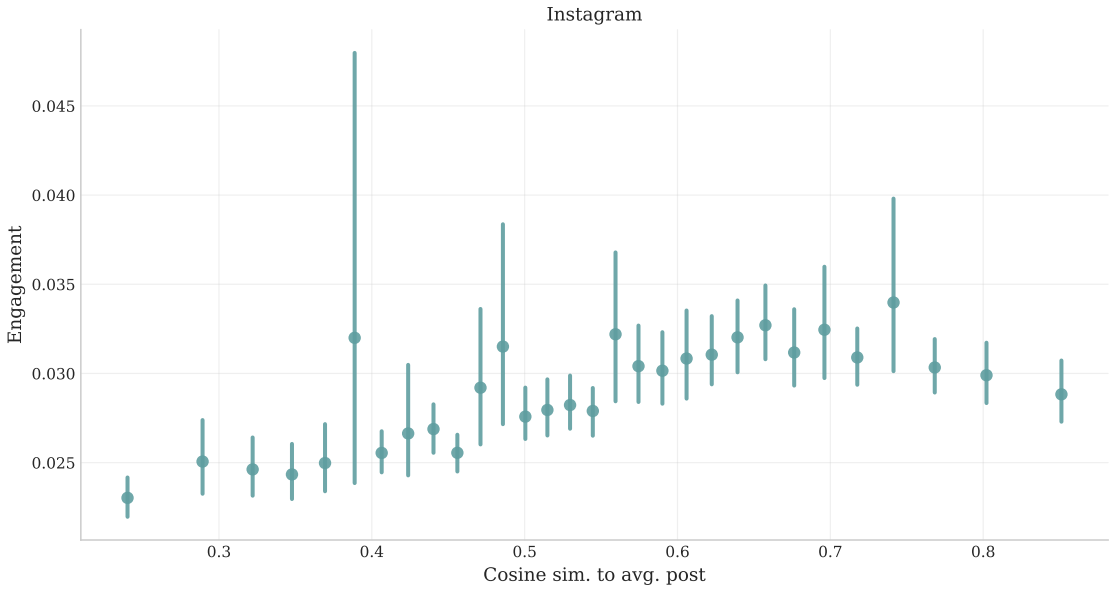


follower-brand fit and the performance of an influencer marketing campaign, although they calculate fit in a different way. I test this in my data by plotting engagement (likes plus comments divided by followers) and match value for sponsored Instagram posts (Figure C.2). The relationship does seem to have an inverted U shape. The sponsored posts that see the highest engagement are not the ones most similar to the influencer's typical content. My measure of fit is different from Leung et al. (2022). They measure the overlap between the brand's category (e.g. beauty) and the categories in which the audience is interested.

With this measure, crowding out is one reason for an inverted U relationship. If the audience is interested in beauty and travel, beauty/travel brands will face fierce competition since many similar brands will offer to sponsor the influencer. A beauty/tech brand can still appeal to the audience while avoiding competition.

My text similarity-based measure exhibits a similar pattern for different reasons. Sponsored posts very similar to the influencer's average organic post might seem to followers like an attempt to conceal the sponsorship. Alternatively, they might not stand out from the influencer's typical content, while a sponsored

Figure C.2: Engagement vs similarity to average organic post



post that is very different attracts more attention from followers. The pattern could also reflect different behavior by follower count. Some of the largest influencers have a set "formula" for their content, so their sponsored posts will tend to have very high similarity scores. Larger influencers also tend to see lower engagement than mid-sized influencers.

Appendix D Transition function

D.1 Unit root problem

Follower count in week $t+1$ is typically very similar to follower count in week t, so regressing $\log f_{it+1}$ on $\log f_{it}$ would yield a coefficient close to one. Such a regression does not produce reliable estimates of the other coefficients. Most formal unit root tests require a balanced panel, so I create a balanced version of my panel running from June 30th, 2020 to June 30th, 2023. I exclude influencers who have any weeks with missing follower count data in this period; the resulting panel has 1,130 influencers. I apply a test that combines hypothesis tests for individual influencers. I can reject the null hypothesis of having a unit root for only 111 of the influencers, so it is possible that many of the follower count time series do have unit roots. This justifies my choice to use the *change* in follower count as the dependent variable in Equation 5.1.

D.2 Alternative specifications for the transition function

Here, I estimate different versions of the transition equation (3.4) to explore the robustness of the finding that organic and sponsored posts have similar effects on follower count growth.

D.2.1 Fraction of sponsored posts

Followers might care not about the total quantity of advertising, but about its pervasiveness throughout the influencer's content. Influencers who are primarily focused on commercializing their content might see slower growth. Anecdotally, an influencer marketing firm told me they recommend influencers keep their fraction of sponsored content below a threshold. In this case the influencer's follower growth depends on the total number of posts they produce and on the fraction of those that are sponsored. To determine this effect I estimate the following equation:

$$\log f_{it+1} - \log f_{it} = \tau_p(o_{it} + s_{it}) + \tau_r \frac{s_{it}}{o_{it} + s_{it}} + \tau_\ell \ell_{it} + \epsilon_{it} \tag{D.1}$$

Table D.1 shows the result. Increasing the fraction of sponsored posts reduces follower growth, but the effect is small. An influencer who grows from 100,000 followers to 101,000 followers with zero sponsored content would grow to about 100,894 followers if 100% of their content were sponsored that period, a loss of 106 followers.

D.2.2 Cumulative effects in the transition function

The negative effects of sponsored content might build up over time. Followers might be willing to tolerate the occasional week of mostly sponsored posts, but if the influencer produces heavily commercialized content week after week, their growth could slow. Other cumulative effects might impact the probability that a user discovers the influencer. Influencers with more total posts or higher quality past posts could appear more often in searches and on algorithmically curated discovery pages, in which case they would see faster follower growth. I calculate an influencer's total "stock" of posts P_T at time T as

$$P_T = \sum_{t=0}^{T-1} \beta^{T-t}(o_t + s_t) \tag{D.2}$$

Table D.1: Empirical transition function including authenticity

	Change log followers	Change log followers	Change log followers
# Posts	0.00024***	0.00024***	0.00018***
	(7.3251e-06)	(1e-05)	(1e-05)
Posted * Frac. sponsored		-0.00030***	-0.00046***
		(8e-05)	(8e-05)
Posted * Log likes			0.00029***
			(2e-05)
N	301,392	301,392	301,392
Creator FE	Yes	Yes	Yes
Date FE	Yes	Yes	Yes
Category FE	Yes	Yes	Yes
R2	0.00354	0.00359	0.00457

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

where $\beta = 0.99$ is the discount factor. I calculate a time-discounted stock because Instagram's discovery algorithm seems to put some weight on recency. The details of the algorithm are not public, but my own Explore page shows exclusively posts made this year. Viral trends also arise and die out quickly on social media, so older content quickly loses relevancy. Discounting old posts when calculating an influencer's post stock accounts for these effects. To measure the past performance of an influencer's content, I calculate the average number of likes on all posts prior to the current period. Table D.2 adds these cumulative effects to the transition function regression. The cumulative variables both negatively affect follower growth, probably because they proxy for influencer size (follower count) and time on the platform. Influencers who started making content earlier will likely have a larger stock of posts, and influencers with more followers tend to get more likes. If a particular content category has a "carrying capacity", or a maximum number of users interested in the subject, then influencers in that category will see slower growth as they approach the upper bound. This could explain the negative effects of post stock and cumulative average likes.

D.2.3 Persistence

Reverse causality might cause problems in the transition function regression. If influencers who are growing more quickly receive more offers to make sponsored content, then growth causes an increase in the number of sponsored posts rather than vice versa. For a brand, fast-growing influencers are particularly valuable. The price of a sponsored post in the current period is relatively small, but the influencer will command a much larger following in the future, and the brand's advertisement will be distributed to that larger audience. The dependent variable in my transition function regressions is the change in log followers from week t to week $t+1$, but a brand does not know the influencer's future follower count. Instead, it might forecast based on the influencer's growth in the previous period. To account for this possibility, Table D.3 introduces lagged change in follower count to explain follower growth. The first column is my baseline specification. The second column includes the change in log followers from week $t-1$ to week t, and the third column adds the change in followers from week $t-2$ to week $t-1$. Including past follower growth increases the R^2 substantially, and including an additional lag yields a slightly larger increase. Given the changes in the R^2,

Table D.2: Empirical transition function with cumulative effects

	Change log followers	Change log followers	Change log followers	Change log followers
# Organic posts	0.00019***	0.00029***	0.00015***	0.00023***
	(1e-05)	(1e-05)	(0.00001)	(0.00001)
# Sponsored posts	0.00010***	0.00020***	0.00011***	0.00017***
	(2e-05)	(2e-05)	(0.00002)	(0.00002)
Posted * Log likes	0.00028***	0.00033***	0.00063***	0.00063***
	(2e-05)	(2e-05)	(0.00002)	(0.00002)
Post stock		-0.00001***		-0.00001***
		(0e+00)		(0.00000)
Log avg. likes			-0.01003***	-0.00905***
			(0.00012)	(0.00013)
N	301,392	301,392	295,174	295,174
Creator FE	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes
Category FE	Yes	Yes	Yes	Yes
R2	0.00451	0.0146	0.0268	0.0319

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

a single lag captures most of the effect of past follower growth on current follower growth.

I think of past follower growth as a control for the frequency with which the influencer receives offers to make sponsored content. Influencers who grow more quickly receive more offers. When I include this control, the coefficients on organic and sponsored posts shrink but remain positive and are of the same order of magnitude as in my baseline specification. Given the already-small coefficients, these changes will not result in a significant change in follower growth and should not substantially affect my results.

D.2.4 Match value in the transition function

The alignment between the brands that sponsor an influencer and the influencer's audience affects engagement (Figure C.2), so it might also affect follower growth. When sponsored posts are not too far from the influencer's typical content, they might not have a large impact on growth. Table D.4 adds my measure of follower-brand fit to the transition function regression (I describe the measure in Section C). The negative coefficient on match value might reflect the inverted U relationship between engagement and follower-brand fit: sponsored posts that are too similar to the influencer's typical content perform worse and therefore generate less follower growth.

D.2.5 Alternative classifications of sponsored posts

Table D.5 summarizes my estimates of the empirical transition function using different methods to classify sponsored posts. The coefficient on the number of organic posts is qualitatively very similar across classification methods. It shrinks in the rightmost column most likely because Ershov and Mitchell (2020) use a broad definition of sponsorship that probably includes some organic posts (because of classification words like "have"). Classifying sponsored posts using only the platform's official "paid partnership" label is the

Table D.3: Empirical transition function with lagged change in followers

	Change log followers	Change log followers	Change log followers
# Organic posts	0.00019***	0.00012***	0.00010***
	(1e-05)	(0.00001)	(0.00001)
# Sponsored posts	0.00010***	0.00005**	0.00004*
	(2e-05)	(0.00002)	(0.00002)
Posted * Log likes	0.00028***	0.00020***	0.00019***
	(2e-05)	(0.00002)	(0.00002)
Change log followers (1 lag)		0.33673***	0.30688***
		(0.00166)	(0.00180)
Change log followers (2 lags)			0.11501***
			(0.00175)
N	301,392	299,221	297,080
Creator FE	Yes	Yes	Yes
Date FE	Yes	Yes	Yes
Category FE	Yes	Yes	Yes
R2	0.00451	0.125	0.141

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

only method that generates a negative coefficient on the number of sponsored posts. These are the most obviously sponsored, so consumers react the most negatively to them. The coefficients on the number of sponsored posts are qualitatively quite similar and typically smaller than the coefficient on the number of organic posts. Although my original classification is neither perfect nor the only option, the magnitudes of the coefficients and the difference between them seem broadly correct, so using a different method would likely not change the results of my structural estimation.

Table D.4: Empirical transition function with match value

	Change log followers	Change log followers
# Organic posts	0.00019***	0.00019***
	(1e-05)	(1e-05)
# Sponsored posts	0.00010***	0.00017***
	(2e-05)	(3e-05)
Posted * Log likes	0.00028***	0.00029***
	(2e-05)	(2e-05)
Posted * Match value		-0.00035***
		(1e-04)
N	301,392	301,392
Creator FE	Yes	Yes
Date FE	Yes	Yes
Category FE	Yes	Yes
R2	0.00451	0.00456

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table D.5: Empirical transition function with sponsorship classifications from Ershov and Mitchell (2020)

	Change fol.	Change fol.	Change fol.	Change fol.	Change fol.	Change fol.	Change fol.	Change fol.
# Organic posts	0.00019***	0.00018***	0.00019***	0.00015***	0.00019***	0.00018***	0.00018***	0.00009***
	(1e-05)	(1e-05)	(1e-05)	(1e-05)	(1e-05)	(1e-05)	(1e-05)	(1e-05)
# Sponsored posts	0.00010***							
	(2e-05)							
Posted * Log likes	0.00028***	0.00030***	0.00028***	0.00029***	0.00029***	0.00030***	0.00029***	0.00028***
	(2e-05)	(2e-05)	(2e-05)	(2e-05)	(2e-05)	(2e-05)	(2e-05)	(2e-05)
# Spon. discl. posts		0.00002						
		(2e-05)						
# Spon. undiscl. posts			0.00023***					
			(3e-05)					
# Posts w/ mention				0.00007***				
				(1e-05)				
# Posts w/ FYPM mention					0.00015***			
					(3e-05)			
# Paid partnership posts						-0.00013**		
						(4e-05)		
# Spon. discl. posts (EM)							0.00008***	
							(2e-05)	
# Spon. posts (EM)								0.00013***
								(1e-05)
Creator FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Category FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	301,392	301,392	301,392	301,392	301,392	301,392	301,392	301,392
R2	0.00451	0.00442	0.00461	0.00453	0.00451	0.00445	0.00447	0.00478

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Appendix E Calculating standard errors

I calculate the standard errors of $\widehat{\theta}_e$ and $\widehat{\theta}_m$ using the general formula

$$\widehat{\text{Var}}(\widehat{\theta}_e, \widehat{\theta}_m) = \frac{1}{I}\left(\widehat{D}^\top \widehat{W} \widehat{D}\right)^{-1} \widehat{D}^\top \widehat{W} \widehat{S} \widehat{W} \widehat{D} \left(\widehat{D}^\top \widehat{W} \widehat{D}\right)^{-1}$$

where D is the gradient matrix of the moment conditions with respect to (θ_e, θ_m), S is the covariance matrix of influencer-level moment contributions, W is the weighting matrix used when estimating $(\widehat{\theta}_e, \widehat{\theta}_m)$, and hats denote the estimates of these matrices. The process to calculate $\widehat{W}$ is described in Section 7.2.2.

I estimate S using its sample analog. Let $g_i(o_{it}, s_{it}, \theta_e, \theta_m)$ be the $2B \times 1$ column vector formed by stacking $g^1, \ldots, g^B$ (Equation 7.1) for a specific influencer i. Define the individual-level moment contribution

$$\widehat{G}_i := \frac{1}{T} \sum_{t=1}^{T} g_i(o_{it}, s_{it}, \widehat{\theta}_e, \widehat{\theta}_m)$$

and let $\overline{G} := \frac{1}{I} \sum_{i=1}^{I} \widehat{G}_i$ be the average of these. Then

$$\widehat{S} := \frac{1}{I} \sum_{i=1}^{I} \left(\widehat{G}_i - \overline{G}\right)\left(\widehat{G}_i - \overline{G}\right)^\top$$

This definition of $\widehat{S}$ allows the moment contributions from a specific influencer to have serial dependence over time. The definition assumes that moment contributions are independent *across* influencers.

Denoting the true parameter values by (θ_e^0, θ_m^0), the matrix D is defined as

$$D := \left.\frac{\partial \mathbb{E}\left[g(o_{it}, s_{it}, \theta_e, \theta_m)\right]}{\partial (\theta_e, \theta_m)^\top}\right|_{(\theta_e, \theta_m) = (\theta_e^0, \theta_m^0)}$$

I estimate D using finite differences, perturbing parameter values by 1%, so

$$\widehat{D} := \left[\frac{g_{IT}(\widehat{\theta}_e + 0.01\widehat{\theta}_e, \widehat{\theta}_m) - g_{IT}(\widehat{\theta}_e - 0.01\widehat{\theta}_e, \widehat{\theta}_m)}{0.02\widehat{\theta}_e}, \frac{g_{IT}(\widehat{\theta}_e, \widehat{\theta}_m + 0.01\widehat{\theta}_m) - g_{IT}(\widehat{\theta}_e, \widehat{\theta}_m - 0.01\widehat{\theta}_m)}{0.02\widehat{\theta}_m}\right]$$

which is a $2B \times 2$ matrix.